

Global FXEM Strategy

Assessing Fair Value in FX: A FEER Approach

We introduce our Fundamental Equilibrium Exchange Rate (FEER) model. FEER estimates of fair value are based on the sustainable external position of an economy – a useful framework at a time of trend changes in FX markets and capital flow uncertainties. We will use the results to anchor our long-term forecasts.

Key Takeaways

- We are often asked about fair value in currencies. Yet estimating fair value in FX is challenging, given the lack of anchor or intrinsic cash flows to discount.
- We introduce our FEER model for 20 currencies, which aims to find the level of the real exchange rate that is consistent with external balance.
- This complements our existing BEER model, both of which will anchor the end point of our FX forecast time horizon (18-24m).
- The output isn't necessarily used to inform our short-term investment recommendations, so we stay short USD even though this model flags USD as moderately undervalued.
- The FEER model is not without limitations and we provide an overview of the pros and cons versus alternative approaches.

We're expanding our FX forecasting toolkit by introducing the FEER (Fundamental Equilibrium Exchange Rate) model, alongside our existing BEER (Behavioral Equilibrium Exchange Rate) framework.

Our FEER framework focuses on the long-run equilibrium rate consistent with sustainable external balances, while our BEER framework translates trends in productivity and terms of trade into currency valuations.

We use these models to anchor the end point of our FX forecasts (18-24 months forward), and the results of our FEER model will be incorporated in our next forecasting process. We note that both these frameworks estimate the fundamental value of currencies in the medium-to-long term and can thus differ from our shorter-term FX views.

The Morgan Stanley FEER model: We've constructed FEER estimates for a total of 20 DM and EM currencies by mapping the relationship between the real effective exchange rates (REER) and sustainable current account balances, using the IMF's external balance assessment norms, other macroeconomic variables, and anchored within an ARDL architecture. This framework provides long-term fair value benchmarks aligned with an external equilibrium.

MORGAN STANLEY & CO. INTERNATIONAL PLC+

James K Lord

Strategist

James.Lord@morganstanley.com

+44 20 7677-3254

MORGAN STANLEY INDIA COMPANY PRIVATE LIMITED+

Nimish M Prabhune

Strategist

Nimish.Prabhune@morganstanley.com

+91 22 6996-1862

MORGAN STANLEY ASIA LIMITED+

Gek Teng Khoo

Strategist

Gek.Teng.Khoo@morganstanley.com

+852 3963-0303

MORGAN STANLEY & CO. LLC

Matthew Hornbach

Strategist

Matthew.Hornbach@morganstanley.com

+1 212 761-1837

MORGAN STANLEY & CO. INTERNATIONAL PLC+

David S. Adams, CFA

Strategist

David.S.Adams@morganstanley.com

+44 20 7425-3518

MORGAN STANLEY & CO. LLC

Andrew M Watrous

Strategist

Andrew.Watrous@morganstanley.com

+1 212 761-5287

Ioana Zamfir

Strategist

Ioana.Zamfir@morganstanley.com

+1 212 761-4012

MORGAN STANLEY MUFG SECURITIES CO., LTD. +

Koichi Sugisaki

Strategist

Koichi.Sugisaki@morganstanleymufg.com

+81 3 6836-8428

MORGAN STANLEY & CO. LLC

Zoe K Strauss

Strategist

Zoe.Strauss@morganstanley.com

+1 212 761-0407

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Arnav Gupta

Strategist

Arnav.Gupta@morganstanley.com

+44 20 7677-0382

David Cueva

Strategist

David.Cueva@morganstanley.com

+44 20 7425-2143

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USD neutral to moderately undervalued: Our latest FEER estimates are for 2Q25 and suggest that the latest USD REER for August 2025 was ~5% below this fair value. This is still in the zone of fair value, given the uncertainty bands involved, and reflects recent USD weakness. TRY, MXN, BRL and PLN stand out as the most overvalued versus USD, with misalignments stretched in the 15-30% range. In contrast, KRW, CNY, JPY and INR appear undervalued by 4-10%.

We present two sets of valuation measures – REER-based and bilateral USD/CCY-based estimates – which largely align, with minor deviations attributable to multilateral FX dynamics: Our FEER estimates are broadly aligned between REER-based and USD/CCY valuations, with only minor deviations due to REER's multilateral nature, which thus reinforces the value of using both perspectives – REER for structural positioning and USD/CCY for tactical execution.

Our FEER analysis reveals that EM currencies correct relatively faster, while G10 misalignments persist, driven by flow dynamics and policy sensitivity: We extend our REER-based methodology to generate forward-looking forecasts from the FEER model by estimating the pace at which currencies converge to fair value. On average, it takes 16-19 quarters for currencies to correct half of their misalignment, but speeds vary. G10 currencies tend to adjust more slowly, supported by deep markets and reserve status, while EM currencies tend to mean-revert faster given their flow-driven dynamics and central bank FX reaction function.

We stay short USD: The neutral-to-moderately cheap valuation of USD our model suggests is not too surprising considering the sharp decline in 1H25 and that the ARDL framework assumes some inertia or persistence in REER movements. The sharp decline in USD seen in 1H25 was thus likely captured as an overshooting, making the consolidation seen in 3Q25 understandable. However, we expect the USD bear market to reassert itself, given the cyclical softness we see in US growth in the near term and the expected pricing of a deeper Fed easing cycle.

Executive Summary

MORGAN STANLEY & CO. INTERNATIONAL PLC+

James K LordJames.Lord@morganstanley.com

+44 20 7677-3254

MORGAN STANLEY INDIA COMPANY PRIVATE LIMITED+

Nimish M PrabhuneNimish.Prabhune@morganstanley.com

+91 22 6996-1862

We introduce our Fundamental Equilibrium Exchange Rate (FEER) model as a complement to the existing BEER (Behavioral Equilibrium Exchange Rate) framework (see [Global Macro Strategy: Assessing Fair Value in FX: A BEER Approach](#), April 6, 2021). While the BEER anchors forecasts in productivity and terms-of-trade dynamics, FEER is designed to estimate the exchange rate level consistent with sustainable external balances. Both models anchor our long-term exchange rate forecasts, and the FEER outputs will be incorporated in our next forecast revision.

FX valuation matters, but is hard: Currencies lack intrinsic cash flows, unlike bonds or equities, making valuation more elusive. FX prices instead reflect a constantly shifting mix of trade flows, capital flows, policy expectations and market positioning. The multitude of participants in FX markets means that exchange rate levels are not just driven by investor expectations, and their flows and misalignments can persist, in turn shaping economic fundamentals and capital flows and influencing monetary policy responses. Recognizing when currencies diverge materially from equilibrium is essential for managing macro and investment risks.

Core features of the FEER model

- **Macroeconomic balance:** FEER integrates both internal balance (full employment with stable inflation) and external balance (sustainable current account positions) and solves for the exchange rate that is consistent with both internal and external balance.
- **Sustainable benchmarks:** By leveraging the IMF's external balance assessment (EBA) current account norms, the model anchors valuations to externally validated sustainability metrics. These benchmarks embed structural factors such as demographics, fiscal stance and institutional quality, making the framework policy-relevant.
- **ARDL architecture:** We have estimated the FEER equation using the Autoregressive Distributed Lag (ARDL) method. The framework attempts to empirically model the long-run equilibrium relationships, by including a dynamic lag structure of both the dependent and independent variables (due to serial correlation).
- **Empirical foundation:** The model regresses real effective exchange rates (REER) on key structural drivers, including the current account, fiscal balance, net foreign asset position, relative productivity and terms of trade. This allows it to capture the long-run relationships between fundamentals and exchange rates.
- **Policy relevance:** FEER highlights when currencies are trading at levels inconsistent with sustainable balances, providing early warning signals for policymakers and investors. It also allows scenario analysis, for example how fiscal consolidation, productivity gains or shifts in terms of trade would influence

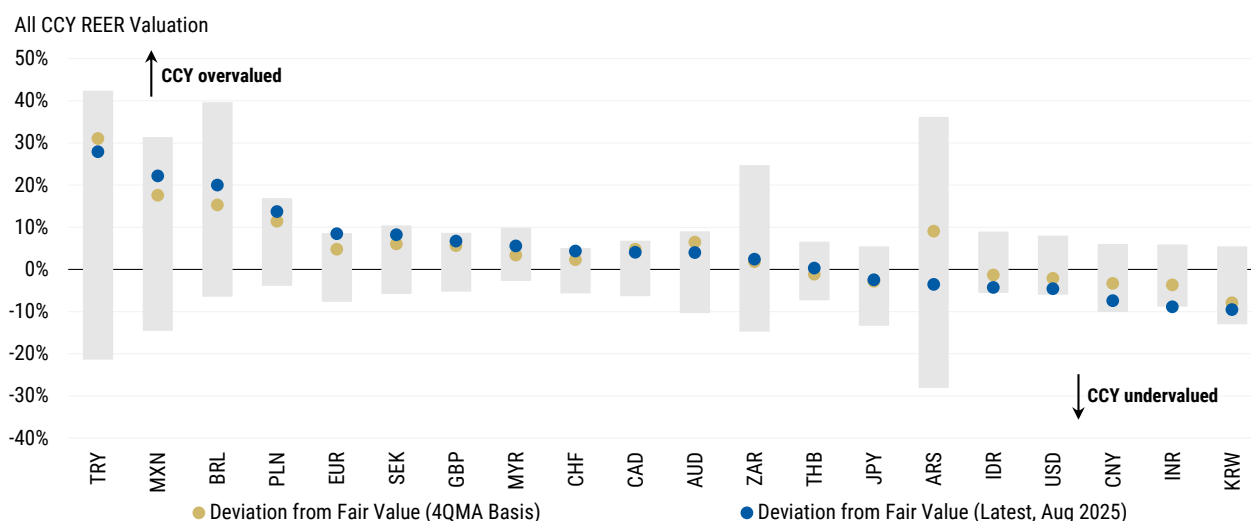
equilibrium exchange rates.

- **Complementary to BEER:** While BEER captures shorter-to-medium-term drivers based on observed fundamentals, FEER offers a structural, longer-term anchor. Together, the two frameworks give a fuller picture of valuation dynamics, distinguishing cyclical misalignments from structural imbalances.
- **Smoothed variables:** All variables are presented on a four-quarter average basis to remove seasonal variations. The estimated FEER values derived in the model, and thus the degree of misalignment, are also initially presented on a four-quarter basis. To aid real-time interpretation, we also present 'mark-to-market' deviations from FEER value for the latest quarter.

Key findings

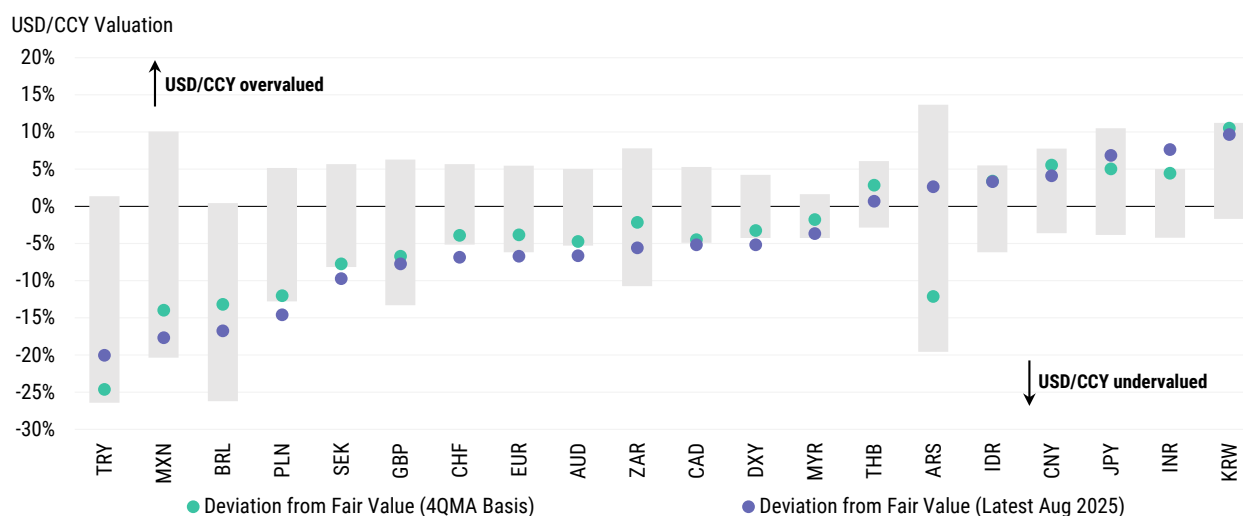
- **USD undervalued:** Despite cyclical strength, the DXY screened as modestly undervalued on a long-term FEER basis in the four quarters to 2Q25. The degree of undervaluation grows when looking at the latest value (August 2025) or USD REER, considering the sharp move lower in USD we have seen year to date.
- **Overvaluation concentrated in LatAm and CEEMEA:** TRY, MXN, BRL and PLN show the largest deviations (15-30% overvaluation).
- **Undervaluation in Asia:** KRW, CNY, JPY and INR appear undervalued by 4-10%.
- **Sharp ARS adjustment:** The recent weakness in ARS moved the currency to neutral territory as of August 2025 after screening as overvalued in the period to 2Q25.
- **Slow convergence in G10 versus faster EM reversion:** G10 currencies appear to stay misaligned for many years, while EM currencies converge faster, potentially due to flow sensitivity and central bank sensitivity to FX movements.

Exhibit 1: Our FEER analysis suggests that TRY, MXN and BRL are overvalued, while KRW, INR and CNY are undervalued on the REER basis



Source: BIS, Morgan Stanley Research estimates; Note: FEER valuation estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

Exhibit 2: FX overvaluation seems concentrated in LatAm/CEE and parts of Europe, while most of Asia screens as undervalued versus USD



Source: Bloomberg, Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

Exhibit 3: Latest G10 and EM FEER valuation profile

Region	REER			Currency Pair	USD/CCY		
	Latest REER (August 2025)	FEER based Estimate	Spot Deviation from Fair Value		Latest FX Cross (August 2025)	FEER based Estimate	Spot Deviation from Fair Value
TRY	98	77	27.9%	USD/TRY	41	51	-20.0%
MXN	128	105	22.2%	USD/MXN	18.7	22.7	-17.7%
BRL	108	90	20.0%	USD/BRL	5.4	6.5	-16.8%
PLN	126	111	13.7%	USD/PLN	3.6	4.3	-14.6%
EUR	105	97	8.5%	USD/SEK	9.5	10.5	-9.7%
SEK	98	91	8.2%	USD/GBP	0.74	0.80	-7.7%
GBP	112	105	6.7%	USD/CHF	0.80	0.86	-6.9%
MYR	101	96	5.6%	USD/EUR	0.86	0.92	-6.7%
CHF	104	99	4.4%	USD/AUD	1.53	1.64	-6.6%
CAD	97	93	4.1%	USD/ZAR	17.7	18.7	-5.6%
AUD	105	101	4.0%	USD/CAD	1.37	1.45	-5.2%
ZAR	104	101	2.5%	DXY	97.8	103.1	-5.2%
THB	98	98	0.3%	USD/MYR	4.2	4.4	-3.7%
JPY	72	74	-2.5%	USD/THB	32.4	32.2	0.7%
ARS	126	131	-3.6%	USD/ARS	1344	1310	2.6%
IDR	95	100	-4.3%	USD/IDR	16490.0	15955.1	3.4%
USD	109	114	-4.6%	USD/CNY	7.13	6.85	4.1%
CNY	87	94	-7.4%	USD/JPY	147.1	137.6	6.9%
INR	96	105	-8.8%	USD/INR	88.21	81.94	7.7%
KRW	91	100	-9.6%	USD/KRW	1389.90	1267.50	9.7%

Source: Bloomberg, BIS, Morgan Stanley Research estimates

Case studies

United States of America (USD): The USD REER was in the zone of neutral as of 2Q25 on our FEER framework. Precisely, the model suggests that USD was undervalued by around 2% at that point, which is modest. The decomposition suggests that USD is overvalued against currencies in Asia.

This result comes despite the US current account balance remaining wider than what the IMF EBA considers to be sustainable, which would indicate that USD is overvalued.

The reason for this is found in our ARDL methodology, which by design uses lagged values of the REER as explanatory variables of the FEER estimate, given serial correlation in REER indices.

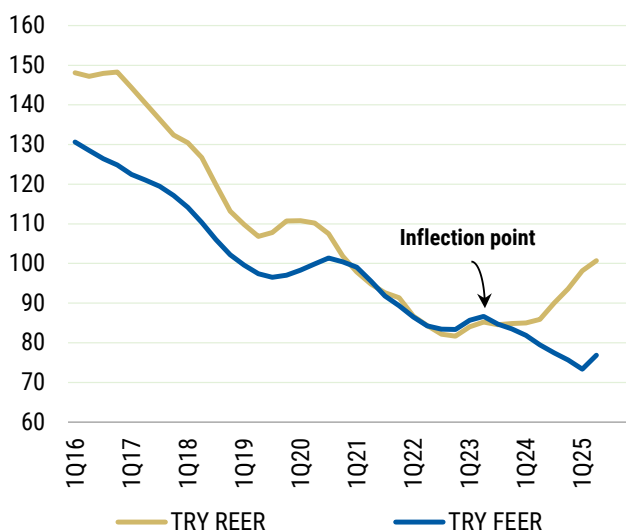
Traditionally, lagged values of the REER are not viewed as fundamental drivers in the same way as variables such as the current account, terms of trade or relative productivity. Instead, they can be seen as capturing inertia or persistence in REER movements. Also, including lagged REER terms allows the model to account for the fact that exchange rates may deviate from their fundamental drivers due to market frictions or temporary shocks.

In other words, the current account variable alone would suggest that USD remains overvalued. However, since a portion of the FEER-based fair value depends on lagged REER values and USD trended higher in 2024, this biases the fair value estimate higher, even as the current account has widened. This effect will likely moderate in future updates considering the move lower in USD in 1H25.

Ultimately USD should converge to levels consistent with what the external balance would suggest (and our model suggests that the current account position has a significant effect over the long term), but the ARDL specification assumes a bumpy and gradual convergence process.

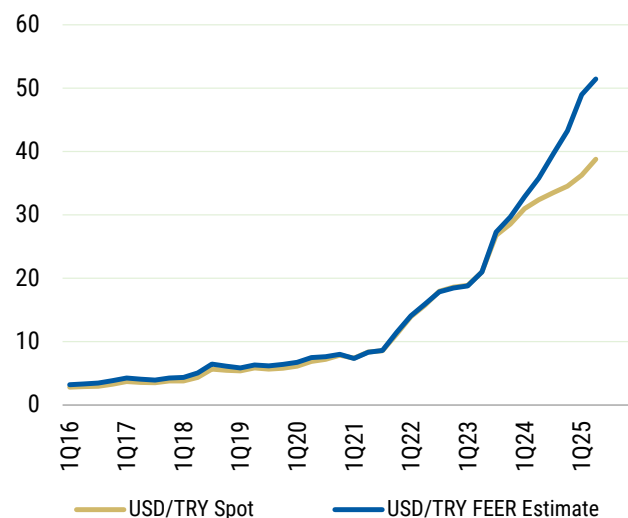
Turkey (TRY): Persistent overvaluation illustrates the limitations of FEER when inflation spirals and policy credibility is reduced. From 2021, the CBT cut rates despite surging inflation, pushing real rates deep into negative territory. Inflation exceeded 85%Y in 2022, capital fled and credibility was impacted. FX-protected deposits and heavy interventions further distorted adjustment, leaving TRY's depreciation far outpacing FEER-implied fair value. The Turkish case underscores how a reduction in institutional strength and heterodox policy can break the link between fundamentals and FX.

Exhibit 4: Turkey REER



Source: BIS, Morgan Stanley Research estimates

Exhibit 5: USD/TRY

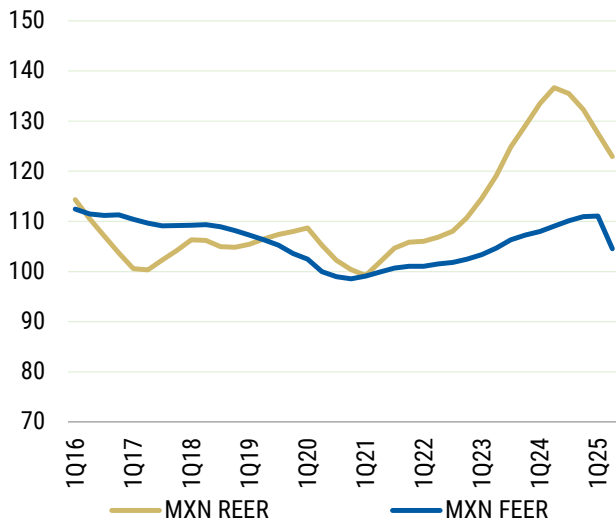


Source: Bloomberg, Morgan Stanley Research estimates

Mexico (MXN): The peso appears persistently rich versus FEER, supported by structural and cyclical forces. Banxico's early aggressive hiking produced some of EM's highest real rates, drawing large inflows. Record remittances and a strong tourism recovery boosted

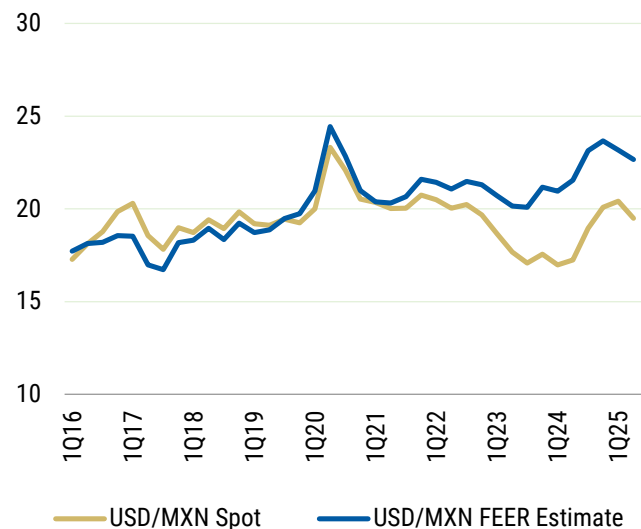
FX receipts further. Beyond flows, nearshoring and the USMCA framework drove sustained FDI, re-rating Mexico's external balance. As a result, MXN trades well above FEER-implied fair value, suggesting that the framework underestimates how structural shifts like supply chain realignment can redefine equilibrium.

Exhibit 6: MXN REER



Source: BIS, Morgan Stanley Research estimates

Exhibit 7: USD/MXN



Source: Bloomberg, Morgan Stanley Research estimates

Limitations and risks

The FEER model is best viewed as a strategic anchor, not a tactical signal for short-term trading. It is sensitive to IMF EBA assumptions and model specification. Policymakers' discretionary choices, especially in managed FX regimes, can sustain deviations far longer than fundamentals imply.

Many FX fair value models can also struggle to incorporate structural changes occurring in real time to produce a forward-looking estimate of where fundamentals are heading.

For example, the IMF EBA assessments include "institutional quality" as one of several variables to explain what an economy's sustainable current account position might be, finding that "a country at the 75th percentile of the institutional quality distribution would have, all else equal, a 0.5 percentage points of GDP lower current account balance".

Investors are currently assessing whether US institutional strength is eroding, and those assessments are being reflected in market prices, including USD, in real time. If there is a change in fundamentals under way there, resulting in a lower FEER estimate for USD, investors will try to price those changes in to some extent via a weaker USD before they are reflected in lower "institutional quality" and sustainable current account position estimates, potentially making USD appear erroneously undervalued.

Why FX Valuation Matters, but Is Difficult

Foreign exchange valuation remains one of the most complex areas in macro strategy:

Unlike equities or fixed income, currencies lack intrinsic cash flows or a clear terminal value, making traditional discounted cash flow or spread-based valuation frameworks inapplicable. Instead, FX prices reflect a constantly shifting equilibrium between macroeconomic fundamentals, policy differentials, capital flows and market positioning. This complexity means that valuation is not an absolute anchor but rather a moving benchmark, one that provides directional context rather than point estimates.

FX valuation demands a nuanced approach, as exchange rates reflect a complex interplay of macroeconomic factors and global capital flows beyond traditional asset pricing models: Exchange rate misalignments can influence trade competitiveness, current account dynamics, capital flows and ultimately monetary policy. Yet measuring 'fair value' is inherently challenging because currencies are simultaneously relative prices (one currency against another) and global assets subject to cross-border portfolio flows. Unlike equities, where earnings expectations dominate, or fixed income, where forward rate pricing is explicit, FX is influenced by a broader and noisier set of drivers, some of them backward looking, including growth differentials, inflation expectations, fiscal and current account balances, terms of trade and risk sentiment.

Furthermore, unlike single-country asset classes, FX valuations require a consistent cross-market lens: A currency may appear undervalued against one trading partner but overvalued against another, depending on macro fundamentals and policy anchors in both economies. This compounds the difficulty of establishing a single 'fair value' of a currency. In practice, valuation frameworks – whether based on purchasing power parity, balance of payments sustainability or financial equilibrium – provide estimates that are highly model dependent and sensitive to underlying assumptions.

Moreover, FX valuation is inherently forward looking but rarely predictive in isolation: Market participants often use valuation as a sanity check, as an input to broader macro views rather than a stand-alone catalyst. A currency deemed undervalued may remain so for extended periods if offset by structural deficits, political uncertainty or divergent monetary policy paths. Conversely, overvalued currencies can persist when supported by strong capital inflows or reserve currency status. This disconnect between valuation and price action underscores the importance of integrating valuation with sentiment, positioning and liquidity dynamics.

In addition, FX markets are uniquely reflexive: Expectations about future policy or macro conditions can influence capital flows today, which in turn affect exchange rates and feed back into those very expectations. This feedback loop makes FX valuation both a mirror and a map, reflecting current macro conditions while shaping future ones. As such, valuation frameworks must be dynamic, incorporating not just static fundamentals but also evolving narratives and regime shifts.

Ultimately, while FX valuation may lack the precision of equity multiples or bond spreads, it remains a vital tool for framing macro risks and opportunities: In a world of shifting growth trajectories and policy divergence, understanding where currencies sit relative to their fundamental anchors, even if those anchors are moving, is essential.

The Morgan Stanley FEER Model

FEER framework

We start by laying the theoretical foundations of the FEER framework: The FEER concept hinges on the notion of macroeconomic balance, encompassing both internal and external balance dimensions and the exchange rate that achieves internal and external balance, simultaneously is defined as the Fundamental Equilibrium Exchange Rate (FEER).

Internal balance is attained when the economy operates at full employment with stable prices (NAIRU). NAIRU stands for non-accelerating inflation rate of unemployment and represents the unemployment rate consistent with stable inflation.

External balance is characterized by a balance of payments position that is sustainable over the medium term, ensuring external debt sustainability.

The approach is focused on determining exchange rates based on specific economic conditions, abstracting from short-term cyclical fluctuations and temporary factors, and concentrating on economic fundamentals which are likely to persist over the medium term. Hence, the FEER exchange rate measure is a **normative one**, serving as a method for calculating a real equilibrium exchange rate consistent with medium-term macroeconomic equilibrium.

The framework is not, in itself, a theory of exchange rate determination. Rather, it serves as a benchmark for where exchange rates should trade, rather than an explanation for why they trade where they do. Yet, it implicitly assumes that the actual REER will not diverge permanently from this benchmark; instead, current account imbalances will drive eventual convergence.

Because it is difficult to identify the level of potential output, it is often assumed that the adjustment process assures internal balance is maintained when external balance is achieved. The external balance component of the model is usually determined by assuming that the savings and investment balance is in some sense sustainable. However, it does not imply that there will be a full stock-flow equilibrium, as that could take decades to achieve.

Given this backdrop, the FEER problem gets reduced to a two-step procedure:

1) Construct a reduced form equation estimating REER as a function of current account balance and other fundamental macroeconomic variables.

$$REER = f(CAB, X_i)$$

2) Impose the condition of macroeconomic balance and solve for the equilibrium REER that achieves it.

$$FEER = f(CAB^*, X_i)$$

where CAB^* is the sustainable current account balance.

Our approach differs somewhat from the original methodology outlined by Williamson (1994), which models the current account balance as a function of the REER to derive the REER adjustment needed to achieve external balance.

However, in practice, this method is not feasible for all economies, as the inversion can yield REER estimates that are overly sensitive to model coefficients, sometimes producing counter-intuitive results due to model mis-specification or omitted variables affecting the current account.

Instead, we adopt an inverse reduced-form specification, modeling the REER as a function of the CAB and structural macroeconomic variables, and compute equilibrium REER levels by substituting in an externally determined estimate of the sustainable current account. While this reverses the conventional causality, it remains conceptually consistent with the FEER framework, given the joint determination of REER and CAB in general equilibrium. This formulation provides a tractable and empirically grounded means of estimating equilibrium REERs, particularly when reliable estimates of sustainable external balances are available.

The notion of a sustainable current account: IMF EBA-CA

The current account can be viewed in terms of saving-investment balances ($CAB=S-I$) and accordingly econometric models are estimated by regressing savings and investment on an array of explanatory variables. Fitted values for investment and saving are then used to derive medium-term values of the current account.

We have used the [IMF's external balance assessment \(EBA\)](#) framework and its published current account balance norms as proxies for sustainable current account levels.

IMF EBA-CA model: The EBA-CA model relates the current account balance, given by the difference between aggregate saving and investment, to a matrix of cyclical and temporal factors, medium-term macroeconomic and structural fundamentals and policy variables ([Exhibit 8](#)).

The estimated EBA-CA model coefficients are used to establish current account norms and their corresponding gaps. The EBA-CA norms are the benchmark levels consistent with the underlying macroeconomic and structural fundamentals at their actual values and with medium-term policies deemed desirable or appropriate by the IMF, once temporary and cyclical factors are excluded.

Hence, the norms represent a normative view of the current account and are not necessarily the fitted values of the model. These norms also depend on desirable medium-term policies deemed appropriate by the IMF staff once cyclical factors are accounted for.

Exhibit 8: IMF EBA regressors

Temporary and Cyclical Factors	Output Gap Commodity Terms of Trade Gap Change in REER
Macroeconomic Fundamentals	Net Foreign Asset Position Output per Worker Expected Real GDP Growth (5 years ahead)
Structural Fundamentals	Demographics Institutional Quality Exhaustible Oil and Natural Gas Reserves
Policy Variables	Fiscal Policy Health Spending FX Intervention interacted with Capital Controls Credit Gap

Source: IMF, Morgan Stanley Research

In the original FEER model specified by Williamson (1994), a currency's fair value is defined as the rate that ensures the current account remains sustainable at full employment. The IMF's EBA current account norms translate this idea into an empirical benchmark by estimating the medium-term external balance justified by structural factors like demographics, fiscal stance and net foreign assets. Thus, EBA norms should, in practice, serve as a credible proxy for the sustainable current account in the FEER framework.

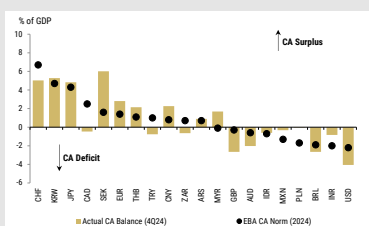
In [Exhibit 9](#) , we show the evolution of each country's EBA gap since 1Q13.

Exhibit 9: Evolution of EBA gaps (as % of GDP)

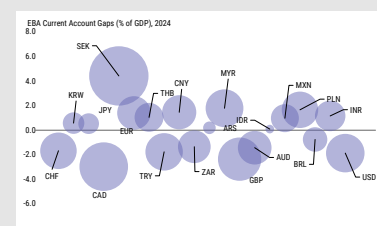
	USD	EUR	JPY	GBP	CAD	SEK	CHF	AUD	CNY	INR	KRW	IDR	MYR	THB	PLN	TRY	ZAR	BRL	MXN	ARS
1013	0.4	0.2	-2.1	-1.9	-3.8	6.8	0.3	-2.7	1.7	-1.0	3.5	-2.8	4.2	-2.7	-1.0	-3.6	-3.0	-1.1	-0.1	
2013	-0.4	0.2	-1.9	-1.9	-3.8	6.1	1.8	-2.7	1.7	-1.0	3.5	-2.8	4.2	-2.7	-1.0	-3.6	-3.0	-1.1	-0.1	
3013	-0.5	0.1	-2.0	-2.7	-3.4	5.9	3.8	-2.6	1.4	-0.1	4.0	-3.1	4.3	-2.7	-0.9	-3.7	-3.4	-1.1	-0.1	
4013	-0.5	0.1	-2.2	-3.3	-3.1	5.5	2.4	-2.1	1.3	1.5	4.2	-2.6	3.5	-1.6	-0.4	-4.0	-3.3	-0.5	0.2	
1014	-0.7	-0.1	-2.6	-3.4	-2.8	5.0	3.2	-2.0	1.1	0.5	4.5	-2.4	4.1	-0.8	-0.7	-3.8	-3.3	0.5	0.4	
2014	-0.8	-0.5	-3.0	-3.6	-2.7	4.7	0.5	-2.0	1.3	3.2	4.3	-2.2	5.2	1.9	-1.3	-3.1	-3.4	0.8	0.3	
3014	-0.9	-0.4	-3.0	-3.9	-2.7	4.7	-0.9	-2.0	1.5	3.0	4.0	-1.8	4.8	1.3	-1.8	-2.8	-3.1	-1.0	0.4	
4014	-1.1	-0.4	-2.6	-4.4	-2.9	4.9	0.4	-2.1	1.8	2.8	4.1	-1.7	3.5	2.0	-2.0	-2.7	-3.4	-1.6	0.5	
1015	-1.2	-0.2	-1.8	-4.9	-3.5	5.0	0.7	-2.4	2.2	2.9	4.5	-1.4	2.9	2.3	-1.7	-2.7	-3.4	-1.8	0.3	
2015	-1.2	-0.2	-1.2	-4.8	-3.7	5.1	1.4	-2.7	2.2	3.0	4.7	-1.0	2.1	2.4	-1.0	-2.7	-2.9	-1.8	0.2	
3015	-1.2	-0.1	-0.7	-4.8	-4.1	4.7	2.8	-2.9	2.2	3.1	5.2	-0.9	1.5	4.9	-0.9	-2.5	-3.1	-1.6	-0.4	
4015	-1.0	-0.2	-0.3	-4.8	-4.2	4.1	1.7	-3.4	2.3	3.1	5.2	-1.0	2.0	5.7	-0.6	-1.9	-3.4	-0.9	-0.6	
1016	-1.0	-0.2	0.0	-4.9	-4.1	3.8	0.3	-3.5	2.1	3.2	5.4	-1.2	1.5	7.4	-0.5	-1.7	-3.8	-0.2	-0.8	
2016	-1.1	0.6	0.2	-5.8	-4.5	2.6	0.2	-3.5	2.2	3.1	5.1	-1.3	1.1	8.7	0.4	-1.6	-3.9	0.4	-0.7	
3016	-1.2	1.0	0.5	-6.6	-4.5	1.5	-0.6	-3.4	2.3	3.1	4.2	-1.3	1.4	10.0	0.3	-1.9	-3.7	0.7	-0.4	
4016	-1.3	1.4	0.7	-6.2	-4.9	0.9	0.8	-2.6	1.8	2.8	3.5	-1.0	1.6	9.8	0.4	-1.7	-3.0	0.7	0.1	
1017	-1.3	1.9	0.8	-5.8	-5.0	0.2	1.1	-2.3	1.7	2.4	2.4	-0.7	1.6	9.4	0.8	-1.8	-2.9	0.9	-1.3	
2017	-1.2	1.5	0.8	-5.4	-4.7	0.7	0.9	-2.0	1.7	1.9	1.6	-0.5	2.3	8.9	0.4	-1.8	-3.0	1.3	0.4	
3017	-1.1	1.9	0.7	-4.8	-4.7	0.8	-0.3	-1.9	1.4	1.9	1.9	-0.4	2.8	9.4	1.2	-2.0	-2.8	1.4	0.5	
4017	-1.0	2.2	1.0	-4.1	-4.9	0.9	-0.6	-2.4	1.9	1.8	1.6	-0.7	2.8	9.4	1.0	-2.7	-2.9	1.6	0.8	
1018	-1.0	2.3	0.9	-4.2	-4.9	0.8	0.3	-2.7	1.4	1.6	1.3	-1.0	3.7	9.2	0.8	-3.3	-3.5	1.2	1.0	
2018	-1.0	2.5	0.9	-4.0	-4.8	0.6	0.5	-2.8	0.9	1.5	1.3	-1.3	3.1	8.4	0.8	-3.5	-3.5	0.8	0.3	
3018	-1.1	2.1	0.5	-4.0	-4.5	1.0	0.5	-2.8	0.7	0.9	1.2	-1.7	2.5	6.3	0.3	-2.4	-3.8	0.2	0.6	
4018	-1.4	1.6	0.1	-4.3	-4.5	1.0	-1.2	-2.4	0.6	0.7	1.1	-2.1	2.4	5.3	0.2	-4.2	-3.8	-0.2	0.2	
1019	-1.5	1.5	0.0	-4.6	-4.6	2.1	-2.9	-2.1	1.0	0.9	1.0	-2.3	2.6	4.7	0.3	-1.8	-3.5	0.3	0.2	
2019	-1.6	1.2	-0.2	-4.2	-4.4	3.4	-3.4	-1.2	1.1	0.8	0.5	-2.3	3.4	4.4	0.5	-3.7	-3.9	-0.5	0.8	
3019	-1.6	1.4	-0.2	-3.8	-4.7	3.9	-3.0	-0.4	1.2	1.1	0.0	-2.3	4.2	5.4	1.0	-4.3	-4.2	-0.8	1.2	
4019	-1.5	1.3	-0.1	-2.5	-4.4	4.1	-2.3	0.1	1.0	1.5	0.0	-2.1	4.0	6.0	1.8	-3.5	-4.0	-1.0	1.7	
1020	-1.5	0.8	-0.2	-1.4	-4.3	4.3	-0.5	0.5	0.5	1.5	0.1	-1.9	3.5	5.7	2.2	-2.6	-3.4	-1.3	1.9	
2020	-1.4	0.9	-0.9	-1.3	-4.4	4.6	-1.7	1.3	0.7	2.6	-0.6	-3.4	2.9	5.4	3.3	0.8	-3.4	-1.0	2.1	
3020	-1.5	0.7	-1.0	-0.9	-4.6	4.1	-1.6	1.7	0.9	3.4	-0.7	-0.6	3.7	4.9	4.0	-2.1	-1.3	0.0	2.5	
4020	-1.7	1.2	-0.9	-2.3	-4.6	4.6	-4.1	2.5	1.1	3.3	-0.3	0.3	4.4	2.8	4.7	-3.4	-0.4	0.1	3.8	
1021	-1.8	1.9	-0.7	-2.0	-3.7	4.7	-5.2	3.2	1.6	2.8	-0.4	0.6	4.5	0.3	4.7	-3.4	0.2	-0.2	3.4	
2021	-1.8	2.5	0.1	-1.0	-3.3	5.2	-3.8	3.5	1.2	2.4	0.3	0.8	4.9	0.5	4.2	-2.4	2.1	0.3	3.8	
3021	-1.8	2.9	0.4	-1.1	-2.6	5.4	-2.1	3.9	1.2	1.6	0.2	1.2	4.5	-2.7	3.1	-0.6	1.6	0.0	2.2	
4021	-1.7	2.8	0.3	0.5	-2.3	5.3	0.5	3.4	1.2	1.1	-0.5	1.3	4.3	-3.1	1.3	0.0	1.5	0.4	1.2	
1022	-1.7	2.6	0.1	-1.0	-2.1	4.9	0.7	2.7	1.3	1.3	-0.7	1.5	3.8	-3.1	0.4	-1.4	1.1	0.5	1.1	
2022	-1.8	1.3	-0.5	-2.1	-1.9	3.8	0.6	3.2	1.4	0.4	-1.1	1.9	3.1	4.1	-0.4	-2.2	-0.2	0.1	0.5	
3022	-1.9	0.0	-1.4	-1.8	-2.3	3.0	0.8	1.0	1.7	-0.3	-2.2	1.8	3.0	-4.7	-0.3	-3.8	-0.8	-0.7	0.2	
4022	-1.8	-0.5	-1.8	-1.7	-2.6	2.8	2.3	1.0	1.6	-0.1	-3.1	1.9	3.5	-4.3	0.0	-4.8	-1.3	-0.2	-1.0	
1023	-1.7	-0.8	-2.4	-1.0	-2.8	3.0	-0.1	1.1	1.4	0.2	-4.4	3.8	3.7	-3.1	1.0	-6.1	-1.6	-0.4	-0.4	
2023	-1.5	-0.3	-1.8	-1.8	-3.4	3.6	-0.2	0.6	1.3	0.5	-4.8	1.5	4.0	-2.2	2.5	-4.7	-1.9	-0.4	-0.1	
3023	-1.4	0.5	-0.9	-2.3	-3.4	4.5	-1.3	0.5	0.8	1.1	-3.8	1.9	3.3	0.0	3.3	-4.1	-1.7	0.3	0.4	
4023	-1.4	1.0	-0.3	-3.1	-2.9	4.7	-1.3	0.3	0.5	1.3	-2.6	0.7	1.9	4.0	4.0	-3.3	-1.7	0.6	1.0	
1024	-1.4	1.2	0.3	-2.6	-2.9	4.4	-0.1	-0.1	0.4	1.4	-1.2	0.2	2.1	0.5	3.7	-2.3	-1.8	0.7	0.9	
2024	-1.5	1.6	0.4	-2.4	-2.8	5.0	0.4	-0.7	0.4	1.4	-0.2	0.2	1.7	1.3	3.2	-2.0	-1.3	0.6	0.9	
3024	-1.8	1.5	0.4	-2.2	-2.7	4.7	-1.1	-0.9	0.9	1.2	0.2	0.1	1.3	1.8	2.3	-1.4	-1.5	0.0	0.9	
4024	-1.8	1.4	0.5	-2.4	-3.0	4.4	-1.7	-1.4	1.5	1.2	0.6	0.1	1.8	1.0	1.7	-1.8	-1.3	0.8	1.0	

Source: IMF, Morgan Stanley Research; Note: Coverage of Argentina in IMF's External Sector Report started in 2018.

[Exhibit 11](#) highlights notable divergences between actual current account balances and those implied by IMF EBA medium-term fundamentals in 2024.

Exhibit 10: IMF EBA current account norms, 2024

Source: IMF, Morgan Stanley Research

Exhibit 11: IMF EBA current account gaps, 2024

Source: IMF, Morgan Stanley Research

For Sweden, Malaysia, Poland, China, the euro area and India, actual balances were **stronger** than fundamentals would suggest. In contrast, Thailand, Mexico, South Korea, Japan, Indonesia, Argentina and Brazil exhibited balances **broadly aligned** with fundamentals.

Meanwhile, Canada, the United Kingdom, the United States, Turkey, Switzerland, Australia and South Africa recorded **weaker** balances relative to medium-term fundamentals.

Model architecture

ARDL regression: We have estimated the FEER equation using the Autoregressive Distributed Lag (ARDL) method. The framework attempts to empirically model the long-run equilibrium relationships, separating them from overlaid short-run dynamics by including a dynamic lag structure of both variables (the regressand and regressors). The framework is also well suited for handling variables that are integrated of different orders $I(0)$ and $I(1)$, and thus does not need pre-testing for unit root conditions.

Additionally, the ARDL framework allows for different optimal lag lengths for each variable instead of a uniform lag for all variables, making it more robust.

The framework has been extensively used in empirical research literature to analyze linkages among macroeconomic variables. Mathematically, the ARDL equation appears as per [Exhibit 12](#) :

Exhibit 12: ARDL equation

$$y_t = \alpha + \sum_{k=1}^p \beta_k y_{t-k} + \sum_{k=0}^{q_1} \beta_{1,k} x_{1,t-k} + \sum_{k=0}^{q_2} \beta_{2,k} x_{2,t-k} + \dots + \sum_{k=0}^{q_n} \beta_{n,k} x_{n,t-k} + \varepsilon_t$$

Source: Morgan Stanley Research

where,

y = dependent variable;

p = optimal lag length of y;

x₁, x₂, ..., x_n = array of 'n' of explanatory variables with optimal lag lengths q₁, q₂, ..., q_n respectively.

Lag selection criteria: To select the optimal lag structure in the ARDL equation, we have used the Bayesian Information Criteria (BIC).

FDL framework: An offshoot of the ARDL framework is the finite distributive lag (FDL) which excludes lags of the dependent variables (autoregressive component) and includes only lagged values of the explanatory variables.

Variable selection and the Morgan Stanley FEER equation

To construct the FEER equation and examine exchange rate misalignment, we have used the following key macroeconomic variables:

- **Terms of trade:** A positive shock to the terms of trade raises export revenues for a given volume, improving the trade balance and thus the current account. Additionally, an improvement in the terms of trade (export prices relative to import prices) typically supports REER appreciation by boosting national income and raising the relative price of non-tradable goods. Conversely, a weaker terms of trade puts pressure on tradable sector competitiveness, which should lead to REER depreciation and can widen current account deficits if not offset by exchange rate adjustments.
- **Fiscal balance:** The fiscal balance and current account are closely linked by the twin-deficit hypothesis, which implies that a higher fiscal deficit, ceteris paribus, tends to reduce national saving and widen the current account deficit. In turn, this external imbalance influences the REER, which may require a depreciation to maintain external sustainability by boosting net exports.

Twin-deficit hypothesis

Expenditure-side GDP identity: $GDP = C + I + G + (X-M)$;

Income-side GDP identity: $GDP = C + S + T$;

$C + S + T = C + I + G + (X-M)$;

$(X-M) = (S - I) + (T - G)$;

Current account balance = private savings + fiscal balance.

- **Net foreign assets:** The current account determines how the net foreign asset stock evolves over time. Sustained current account surpluses build up positive net foreign asset positions, while persistent deficits erode them. Additionally, the net foreign asset position helps to anchor the REER over the medium term as economies with large net external liabilities require a more depreciated REER to generate the trade surpluses to stabilize their external debt.
- **Relative productivity:** Higher relative productivity growth in the tradable sector raises an economy's competitiveness, potentially leading to an equilibrium appreciation of the REER. This is based on the Balassa-Samuelson effect. The idea is that as productivity gains lift incomes and non-tradable prices, the overall price level rises relative to trading partners. Provided demand remains elastic over time,

this tends to strengthen the REER while supporting the current account balance.

The data: We ran individual country ARDL regressions for a set of 20 currencies across the G10 and EM space. To make the data uniform, we carried out the following data adjustments:

1. Magnitude adjustment: All monetary variables were scaled to US\$ billion.
2. Frequency adjustment: All variables were scaled to quarterly frequency as per the required up/down sampling techniques.
3. Since the FEER is grounded in medium-to-long-term structural links between exchange rates and external balances, we have employed some smoothing to filter out short-term noise.
 - a. *The REER series (reported monthly) is first smoothed using a 12-month rolling mean to filter out short-term volatility and then converted to quarterly by averaging within each quarter.*
 - b. *For external balance, the current account balance and nominal GDP are expressed as four-quarter rolling sums before taking their ratio, so that the CAB/GDP series reflects structural trends rather than seasonal or transitory fluctuations.*

These adjustments ensure that the model captures underlying equilibrium medium-term dynamics rather than transient fluctuations.

Morgan Stanley FEER equation: Using the framework delineated in [Exhibit 12](#), we construct the following equation as shown in [Exhibit 13](#):

Exhibit 13: Morgan Stanley FEER model equation

$$REER_t = \alpha + \sum_{k=1}^p \beta_k REER_{t-k} + \sum_{k=0}^{q1} \beta_{1,k} CABGDP_{t-k} + \sum_{k=0}^{q2} \beta_{2,k} TOT_{t-k} + \sum_{k=0}^{q3} \beta_{3,k} FBGDP_{t-k} + \sum_{k=0}^{q4} \beta_{4,k} NFAGDP_{t-k} + \sum_{k=0}^{q5} \beta_{5,k} RELPROD_{t-k} + \varepsilon_t$$

Source: Morgan Stanley Research

where,

REER = real effective exchange rate of the CCY;

CABGDP = current account balance (as % of GDP);

TOT = terms of trade (relative price of exports to imports);

FBGDP = fiscal balance (as % of GDP);

NFAGDP = net foreign assets (as % of GDP);

RELPROD = country's relative productivity relative to its trading partners.

Model results

REER dynamics: AR(2) structure with sign reversal

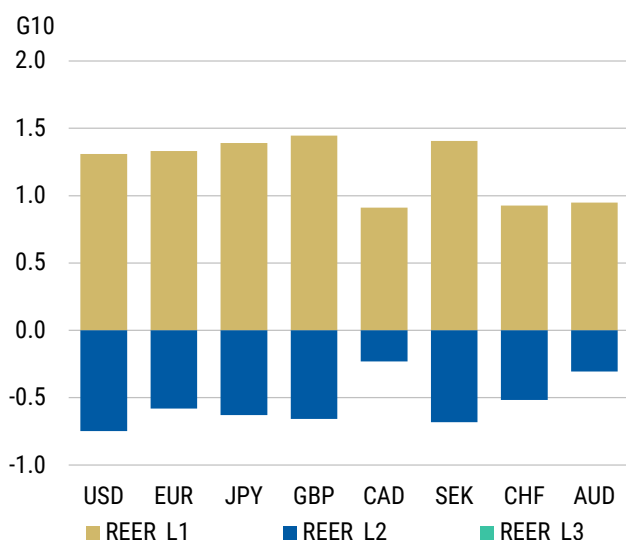
We ran individual country regressions for eight G10 and 12 EM currencies to access the REER relationship.

Across our FEER regressions, we consistently observe an AR(2) process in the estimated REER equations, where the first-order lag enters positively and the second-order lag enters negatively, with both coefficients significant at a 99% confidence level. This pattern is both econometrically intuitive and macroeconomically consistent with the slow-moving adjustment of real exchange rates towards equilibrium ([Exhibit 14](#) and [Exhibit 15](#)).

From an econometric perspective, the ARDL specifications imply that the REER movements exhibit partial adjustment dynamics rather than a simple one-period mean reversion. A positive AR(1) beta indicates persistence in REER deviations (i.e., once the REER moves away from equilibrium, it tends to carry momentum into the next period). However, the negative AR(2) beta suggests that this persistence is not explosive; rather, it introduces a damped oscillatory adjustment path towards equilibrium. **Put differently, currencies tend to overshoot their equilibrium values before gradually correcting back.**

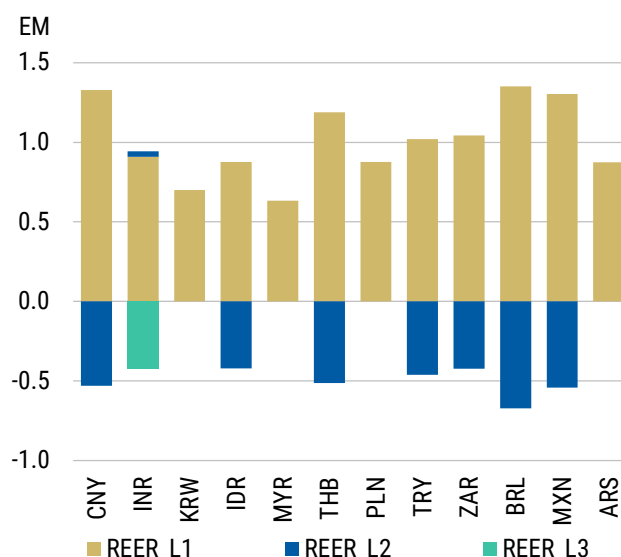
We also observe that the absolute value of the AR(2) beta is less than the AR(1) beta in almost all cases, indicating that the process converges smoothly, but each swing becomes smaller over time until stability is restored.

Exhibit 14: G10: AR betas



Source: Morgan Stanley Research estimates

Exhibit 15: EM: AR betas



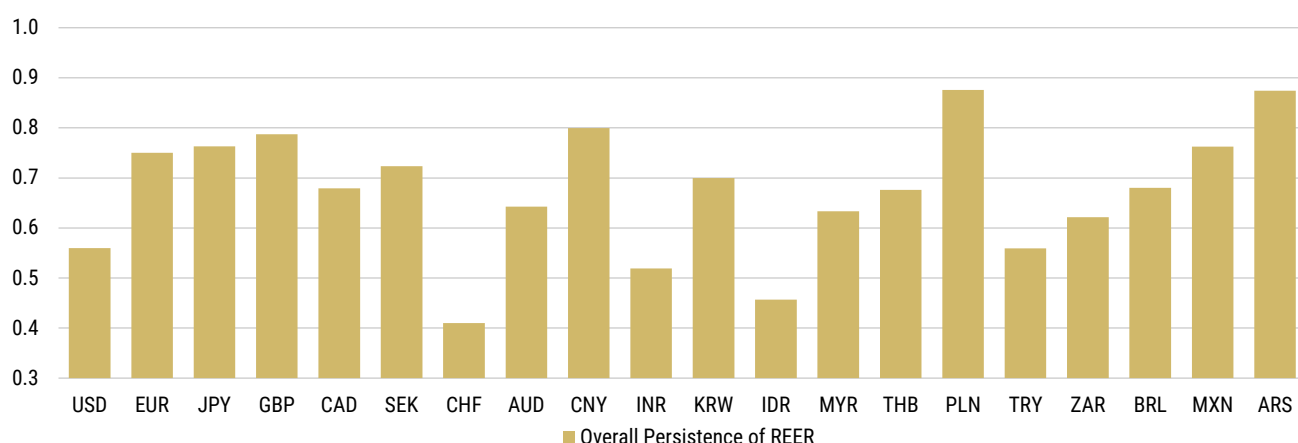
Source: Morgan Stanley Research estimates

From a macroeconomic perspective, this adjustment dynamic reflects how FX markets reconcile short-term capital flows with long-term current account sustainability. A positive AR(1) beta implies that exchange rate shocks (e.g., from policy surprises, portfolio inflows or shifts in risk sentiment) tend to persist initially as markets anchor to recent levels.

The negative AR(2) beta reflects a delayed corrective mechanism, consistent with balance of payments theory, i.e., once short-term flows normalize, the REER adjusts to realign external balances back towards their sustainable path.

Also, across all 20 currencies in our FEER framework, we see that the overall persistence of the REER (measured by the cumulative impact of all lags of the REER) is less than 1. This indicates that REER dynamics are stable and mean-reverting, consistent with our model's assumption that exchange rates ultimately converge toward equilibrium ([Exhibit 16](#)).

Exhibit 16: The cumulative impact of lags (persistence) of the REER indicates a stable and mean-reverting process



Source: Morgan Stanley Research estimates

The observation that this stability condition holds across all currencies provides a statistical validation of the model and reinforces the economic intuition that REER misalignments are temporary and gradually unwind as fundamentals reassert themselves.

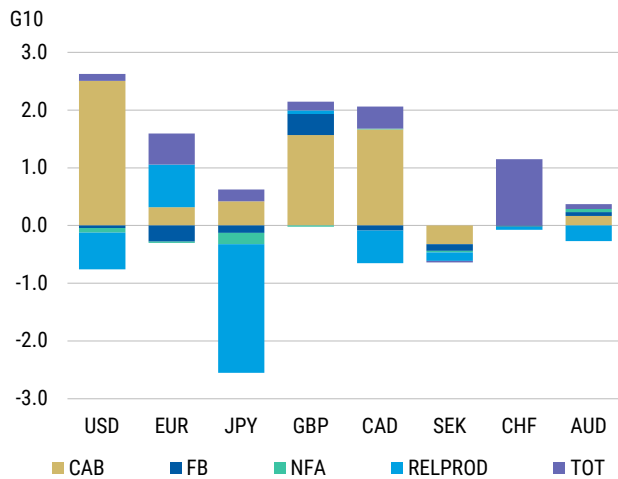
Drivers of REER: Long-run impact dynamics

We next examine the influence of key macroeconomic fundamentals on the REER using long-run betas derived from the ARDL framework: These betas capture the cumulative, persistent impact of each variable once short-term fluctuations have settled, offering a clearer picture of the structural relationship between fundamentals and exchange rates ([Exhibit 17](#), [Exhibit 18](#), [Exhibit 19](#) and [Exhibit 20](#)).

Unlike individual lag terms, which can be noisy and fragmented, long-run betas provide a more stable measure of equilibrium dynamics. This makes them particularly relevant for our FEER framework, where the emphasis is on understanding how sustained shifts in variables, such as the current account balance, fiscal stance or productivity, shape REER trajectories over the medium term.

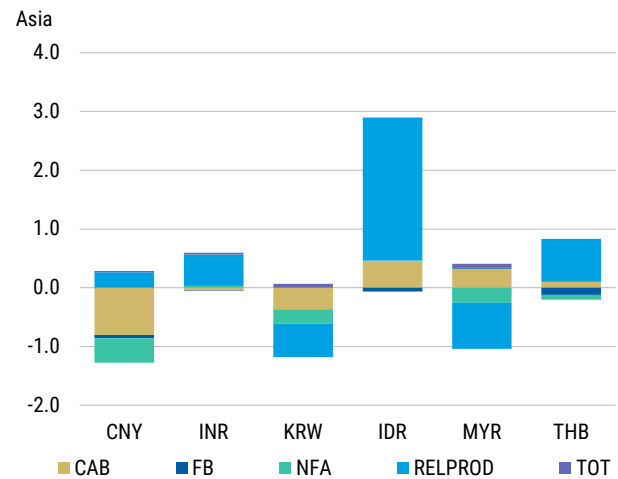
(A detailed summary of the individual ARDL betas and their statistical significance is included in [Appendix 3: Regression Betas](#).)

Exhibit 17: G10



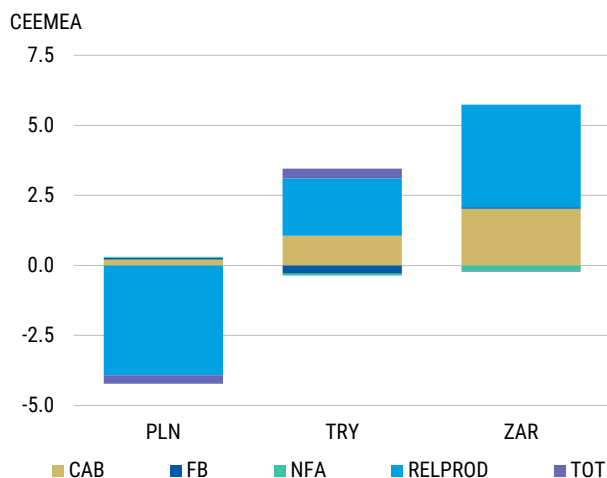
Source: Morgan Stanley Research estimates

Exhibit 18: Asia



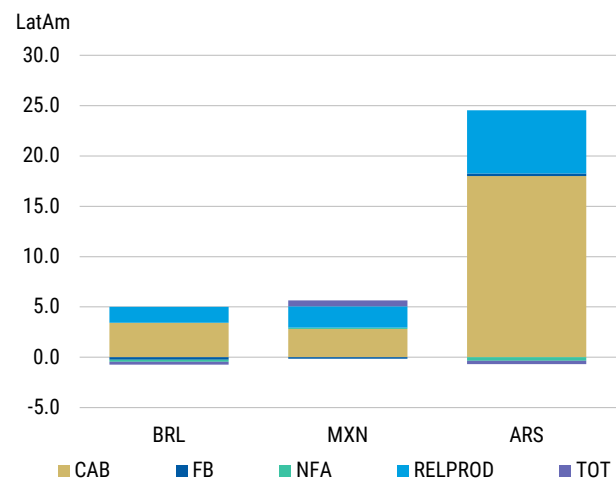
Source: Morgan Stanley Research estimates

Exhibit 19: CEEMEA



Source: Morgan Stanley Research estimates

Exhibit 20: LatAm



Source: Morgan Stanley Research estimates

Current account surpluses and currency strength tend to move together in

equilibrium, even if the mechanics suggest otherwise: Almost 80% of G10 and EM currencies in our sample show a positive sign on the long-run beta of the current account balance, with the exceptions being SEK, CNY, INR and KRW, though the magnitudes are relatively small.

At first glance this observation may seem counter-intuitive as, mechanically, an appreciation in REER should worsen the current account by reducing export competitiveness and boosting imports, implying a negative link.

Yet in practice, current account improvements rarely occur in isolation, but are typically driven by fundamentals like stronger terms of trade, higher productivity, etc. These forces not only lift the current account but also support currency appreciation, producing a positive long-run correlation.

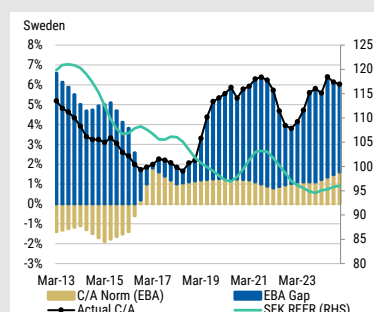
The key distinction is that the accounting identity points to a negative contemporaneous effect, while the equilibrium fundamentals highlight a positive co-movement in the long run. The result is that, in equilibrium, surplus periods and appreciated REERs go hand in hand.

Why SEK, CNY, INR and KRW might be deviating from the pattern?

SEK | Funding currency dynamics: Despite sustained current account surpluses and positive EBA gaps, SEK has not appreciated meaningfully in real effective terms. We think that structural factors such as its status as a low-yielding funding currency, low inflation and credible monetary policy appear to weaken the pass-through from external balances to the exchange rate ([Exhibit 21](#)).

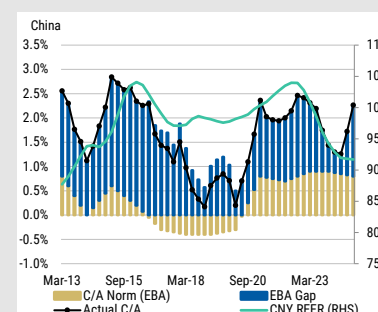
CNY | Policy-managed adjustment: China's managed currency regime indicate that CNY's movements are shaped more by the PBOC's FX reaction function. Even with a sizeable current account surplus, the authorities often limit appreciation to preserve export competitiveness. As a result, the long-run beta is muted or even negative ([Exhibit 22](#)).

Exhibit 21: SEK REER has not appreciated meaningfully...



Source: IMF, BIS, Morgan Stanley Research

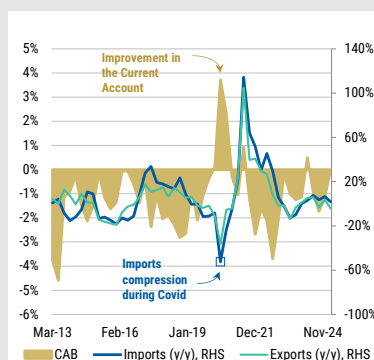
Exhibit 22: ...while CNY REER appreciation is limited



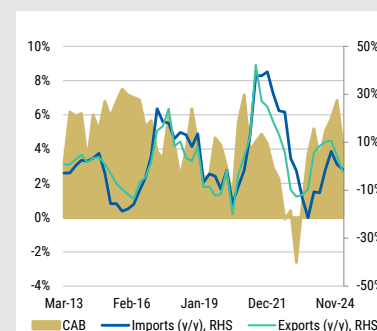
Source: IMF, BIS, Morgan Stanley Research

INR | Import compression led current account improvements: Historically India has run a current account deficit and the current account tends to improve during periods of weak growth, driven more by import compression than export strength. These episodes often coincide with softer capital inflows and weaker growth expectations, leading to INR depreciation even as the external balance appears to improve ([Exhibit 23](#)).

KRW | Cyclical risk-sensitive patterns: Korea's current account often improves during global downturns as imports contract more than exports. However, these same episodes tend to coincide with risk aversion and capital outflows, which weigh on KRW and offset the CA-REER adjustment mechanism ([Exhibit 24](#)).

Exhibit 23: India's CAB improves on import compression...

Source: IMF, Bloomberg, Morgan Stanley Research

Exhibit 24: ...while Korea's CAB improves during global downturns

Source: IMF, Bloomberg, Morgan Stanley Research

In summary, the small negative betas observed for certain economies should be seen as reflecting country-specific idiosyncrasies, rather than evidence of a breakdown in the broader equilibrium relationship between current accounts and exchange rates.

The fiscal channel shows up as weak and inconsistent across currencies, with limited explanatory power for REER: The results were less consistent in the case of fiscal balance, as only six currencies (GBP, AUD, MYR, PLN, ZAR and ARS) out of our G10 and EM sample displayed the theoretically expected sign, while the majority showed weak or opposite relationships with the REER. Moreover, most of the estimated betas were statistically insignificant at a 95% confidence interval and small in absolute magnitude. Overall, the fiscal channel appears weak.

The terms of trade remains a reliable driver of REER, with a few EM outliers: Around 70% of the G10 and EM currencies in our sample exhibited the theoretically expected positive long-run beta, consistent with the view that an improvement in export prices relative to imports supports higher income and, ultimately, an appreciation in the REER.

Within the G10 space, SEK was the only outlier, while among EMs the inconsistency was more evident, with IDR, PLN, BRL, ZAR and ARS showing counter-intuitive signs.

The NFA channel matters at the margin, but its impact is uneven and warrants deeper investigation: The results were less consistent in case of net foreign asset positions, with only six currencies (CAD, AUD, INR, IDR, PLN and MXN) displaying the expected positive long-run beta. While the estimated betas were generally small in absolute magnitude, most were statistically significant at a 95% confidence interval.

This pattern suggests that the NFA channel does influence REER dynamics, but its effect is often muted or masked by other factors. More investigation is needed into analyzing drivers of NFA like the valuation effects, the composition of external liabilities (local versus foreign currency) and the interaction with capital flow sensitivities.

Relative productivity effects on REER are clearer in EM than G10: For relative productivity, the results were mixed. Among the G10 currencies, only EUR and GBP displayed the expected positive long-run beta, while the remaining currencies showed weak or counter-intuitive signs. By contrast, the pattern was more consistent in emerging

markets, where nine out of 12 currencies exhibited the theoretically expected positive long-run beta, with KRW, MYR and PLN standing out as exceptions.

This divergence likely reflects the fact that Balassa-Samuelson effects are more visible in EM, where productivity differentials between tradable and non-tradable sectors are larger and more persistent, while in G10 economies relative productivity gains are narrower and often overshadowed by other drivers such as monetary policy and capital flows.

Despite the mixed signs and significance levels of the long-run betas, we retain all core variables in the FEER model.

From an econometric perspective, excluding variables risks introducing omitted variable bias and weakening the stability of the cointegrating relationship. A reduced model may appear cleaner statistically, but it would be mis-specified if relevant fundamentals are systematically left out.

From a macroeconomic perspective, the FEER framework is intended to capture equilibrium exchange rate dynamics, which requires incorporating the full set of theoretically important drivers. Dropping any of those would distort the structural interpretation of the results and compromise cross-country comparability. By keeping the full set, we preserve the internal consistency of the model, even if individual coefficients must be treated with caution.

Exhibit 25: Average of individual betas of regression inputs

	Average of Individual Betas	
	G10	EM
Current Account Balance	0.8	2.3
Fiscal Balance	0.0	0.0
Net Foreign Asset	0.0	-0.1
Relative Productivity	-0.4	1.2
Terms of Trade	0.3	0.0

Source: Morgan Stanley Research

In the next section, we present our FEER-based currency valuation results.

FX Valuation Results

We provide two kinds of valuation measures: (1) In REER terms; and (2) In USD/CCY terms.

Our valuation results are broadly consistent across REER and their USD/CCY translations, with only minor shifts in ranking reflecting the multilateral structure of the REER.

We compute the exchange rate misvaluations by comparing the actual REER with the estimated FEER. The percentage deviation between the two provides a quantifiable measure of misalignment.

This misalignment serves a dual purpose – it provides a signal of over/undervaluation, and also points to the degree of real exchange rate adjustment needed to bring the current account to its medium-term norm. In that sense, the FEER framework acts as both a valuation tool and a diagnostic of the policy-relevant misalignment that may be necessary to close external imbalances under unchanged fundamentals.

Mark-to-market adjusting the valuations

Since the FEER framework is grounded in medium-to-long-term structural linkages between the REER and external balances, we've used smoothed variables like four-quarter sums of the current account as well as the four-quarter averages of REERs to capture the long-term trend and remove the impact of seasonal variations. As such, our model output indicates whether the REER of any given country was overvalued or undervalued on average over the previous four quarters. While useful, investors will of course want to know whether a currency is overvalued or undervalued today at a specific level, rather than over the past four quarters.

To bridge this gap, we mark our FEER valuations to market by referencing the latest monthly REER value, rather than only the smoothed REER series used in the model. This allows us to highlight the difference between the structural misalignment suggested by fundamentals and the short-term position of the currency in real terms.

This mark-to-market adjustment captures the additional layer of short-term market dynamics that temporarily drive currencies away from their equilibrium path. Put differently, it means that while our model might show a modest over/undervaluation on a smoothed basis, the actual REER can point to larger misalignment and vice versa once short-term volatility is accounted for.

Presenting both perspectives provides a more clear framework: 1) A baseline medium-term anchor from FEER, complemented by 2) A mark-to-market view of prevailing market conditions.

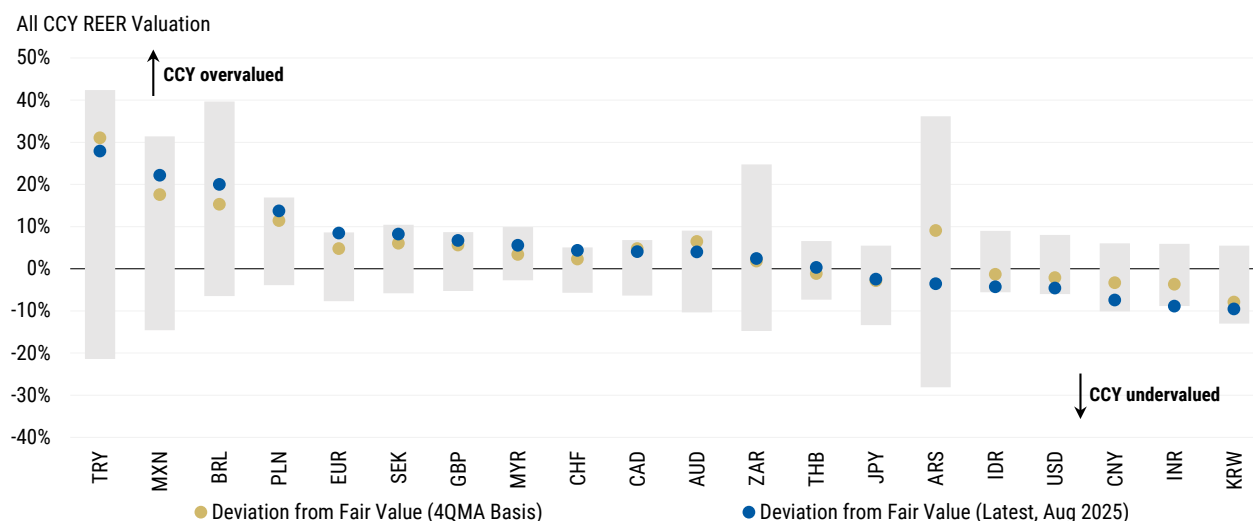
We have updated our valuations to reflect mark-to-market levels as of **end-August 2025**, referencing the actual REER and USD/CCY cross rates.

FEER-based REER valuations

We observe that TRY, MXN and BRL are the most overvalued on an REER basis, while KRW, INR and CNY are at other end of the spectrum.

In 2Q25, our FEER result (presented on a 4QMA basis) indicates that 65% of G10 and EM currencies are trading above fair value, with overvaluation spanning 2-30%. Only 25% currencies in our sample fall into the undervalued category, while the rest remain largely in line with fair value ([Exhibit 26](#)).

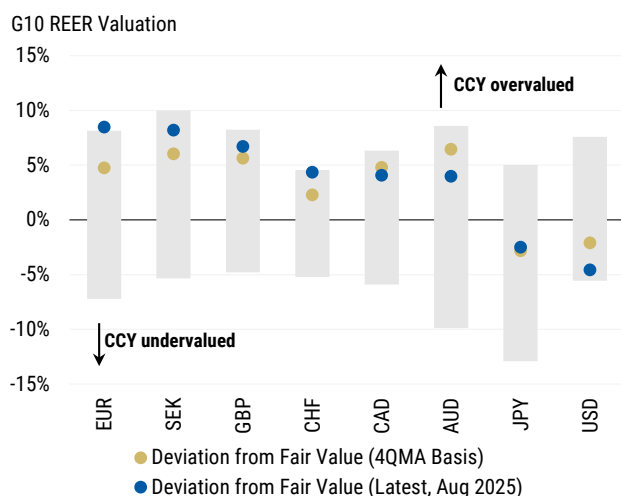
Exhibit 26: TRY and MXN are the most overvalued, while KRW and INR are the most undervalued



Source: BIS, Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

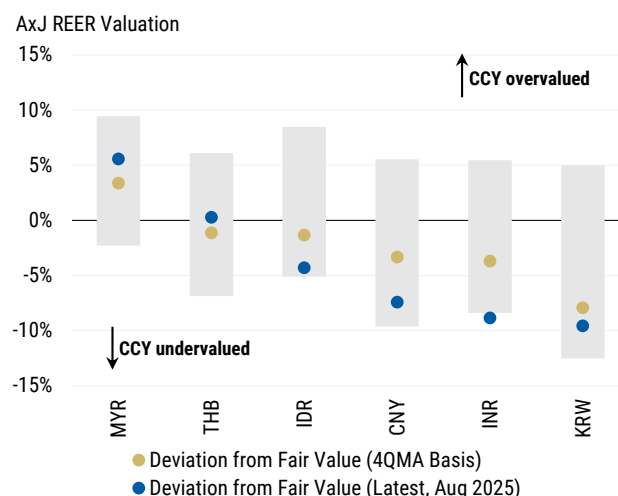
Most G10 currencies continue to trade above fair value, with the mark-to-market adjustment skewing valuations further towards overvaluation. EUR, SEK and GBP appear the most stretched, with REER misalignments of roughly 7-8% as of August 2025, underscoring the relative richness of European FX. CHF, CAD and AUD remain moderately overvalued, while USD and JPY sit in slight undervalued territory. USD's mark-to-market valuation, in particular, highlights recent FX weakness relative to fundamentals ([Exhibit 27](#)).

Exhibit 27: In G10, EUR, SEK and GBP are most overvalued while USD looks a bit undervalued



Source: BIS, Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

Exhibit 28: In Asia, MYR seems overvalued, while the rest are undervalued



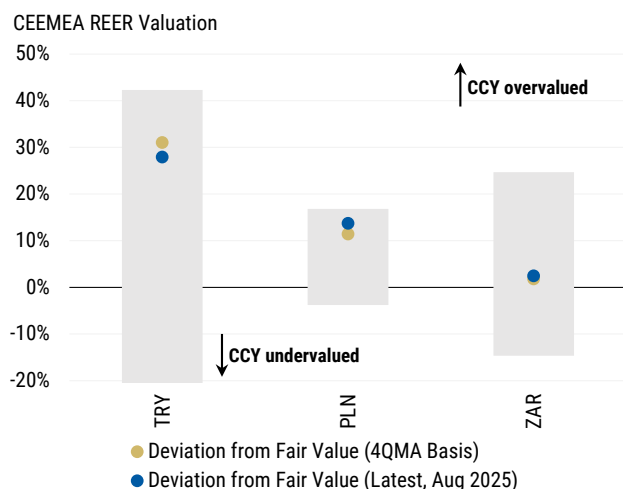
Source: BIS, Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

Within **Asia**, most currencies screen as undervalued, with REER discounts to fair value in the 4-10% range as of August 2025. MYR stands out as the only overvalued currency, trading about 6% above fair value. INR, CNY and IDR show notable mark-to-market adjustments tilted towards undervaluation, while THB has shifted from 4QMA model-based undervaluation to a slight overvaluation in mark-to-market REER terms ([Exhibit 28](#)).

In **CEEMEA**, as of August 2025 TRY and PLN appear significantly overvalued at 28% and 14%, respectively, while ZAR shows a mild overvaluation of 2% ([Exhibit 29](#)).

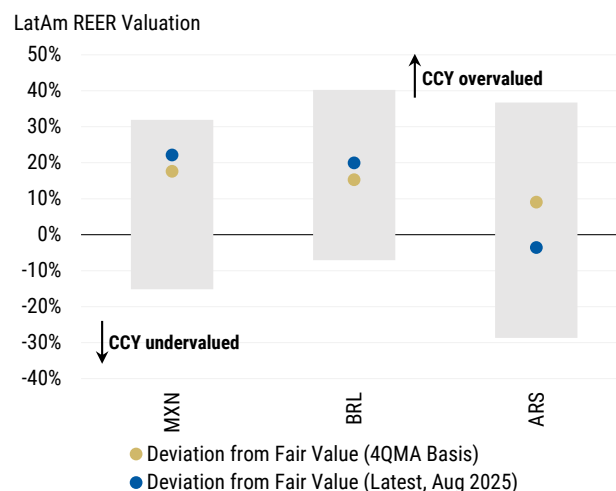
In **LatAm**, MXN and BRL appear overvalued on a 4QMA model basis, with mark-to-market adjustments pushing August 2025 levels even further above equilibrium. ARS is the exception: while it screens as 9% overvalued on a 4QMA model basis to 2Q25, the mark-to-market adjustment shifts it into undervalued territory (around 4%), reflecting recent sell-off pressures ([Exhibit 30](#)).

Exhibit 29: ZAR looks slightly overvalued in CEEMEA...



Source: BIS, Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

Exhibit 30: ...while MXN, BRL look overvalued in LatAm



Source: BIS, Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

REER to USD/CCY conversion

To map multilateral FEER-based REER estimates into bilateral USD/CCY crosses, we employ the method described by Cervero et al. (1999). We start by linearizing the REER indices so that misalignments can be expressed as log-percentage deviations of spot values from fair values. These REER gaps are then decomposed into bilateral currency misalignments by solving the implied system of trade-weighted relationships, which ensures that the set of USD/CCY valuations is internally consistent with the multilateral framework.

The USD consistency constraint: To identify the solution, we impose an additional constraint that the reconstructed USD REER, derived from the bilateral misalignments, matches USD's misalignment from its FEER fair value estimate, thereby providing a coherent anchor for the system. This guarantees cross-rate consistency and preserves the additive structure of exchange rate adjustments across all pairs. **Put differently, adding the USD constraint closes the system and yields a unique solution.**

The method assumes price stability across trading partners over the valuation horizon, so that the adjustment is mapped through currency movements rather than inflation differentials. While this is a simplifying assumption, this assumption delivers a structurally coherent mapping from FEER-based REER misalignments to bilateral USD/CCY valuations.

Specifically, for each country i , assuming prices are unchanged, the log-linearized version of the REER can be worked out as shown in [Exhibit 31](#).

Exhibit 31: Log-linearized REER

$$\begin{aligned} \text{REER}_i &= \prod_{j \neq i} \left(\frac{E_j}{E_i} \right)^{w_{ij}} \\ \Rightarrow \log(\text{REER}_i) &= \sum_{j \neq i} w_{ij} (\log E_j - \log E_i) \\ \Rightarrow \log E_i &= \sum_{j \neq i} w_{ij} \log E_j - \log(\text{REER}_i) \end{aligned}$$

Source: Morgan Stanley Research

where,

E_i = nominal exchange rate for country i (USD/ i), expressed as local currency per USD;

REER_i = real effective exchange rate of country i ;

w_{ij} = trade weight of country j in country i 's REER basket.

The equation above can be written more conveniently for a system of currencies in a vector form as shown in [Exhibit 32](#).

Exhibit 32: Vector form

$$\begin{aligned} m &= W \cdot e - e \\ \Rightarrow (I - W)e &= -m \\ \Rightarrow e &= -(I - W)^{-1} \cdot m \\ \text{s.t. } C \cdot e &= m_{USD} \end{aligned}$$

Source: Morgan Stanley Research

where,

e = vector of bilateral USD/CCY misalignments;

m = vector of REER misalignments;

W = trade-weight matrix;

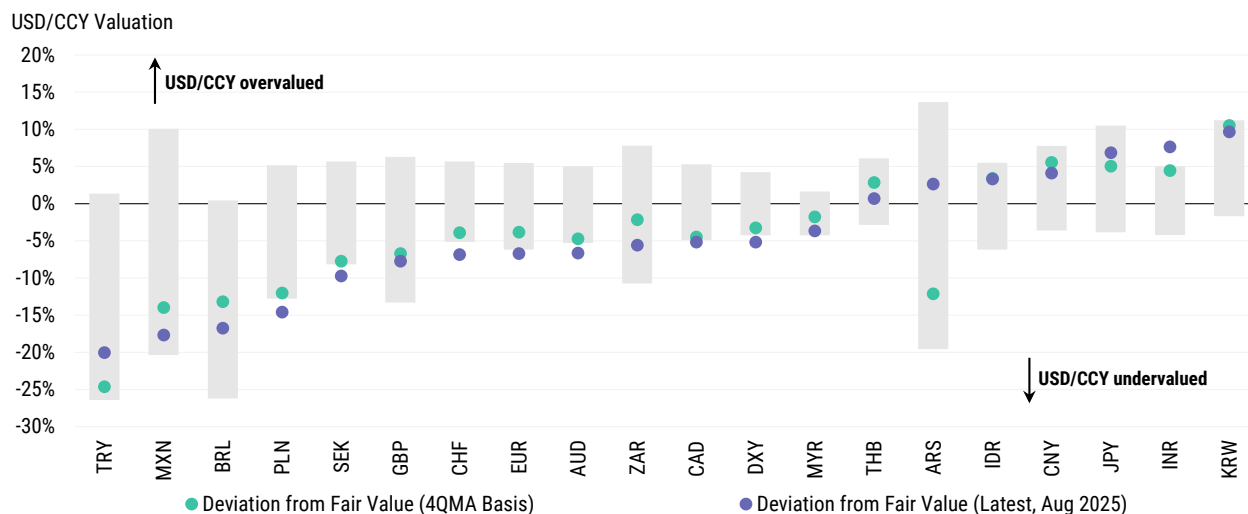
I = Identity matrix;

C = constraint row vector ensuring USD REER consistency.

Our FEER-based analysis of USD/CCY crosses suggests a broad overvaluation bias across most G10 and EM currencies versus USD, with Asia being the key exception, though the magnitudes vary meaningfully.

High-beta EM valuations seem the most stretched, with several currencies in LatAm and CEE screening as notably rich against USD: TRY screens as the most overvalued currency at 20%, while MXN, BRL and PLN also show double-digit misalignments.

Exhibit 33: USD is undervalued against most G10 currencies

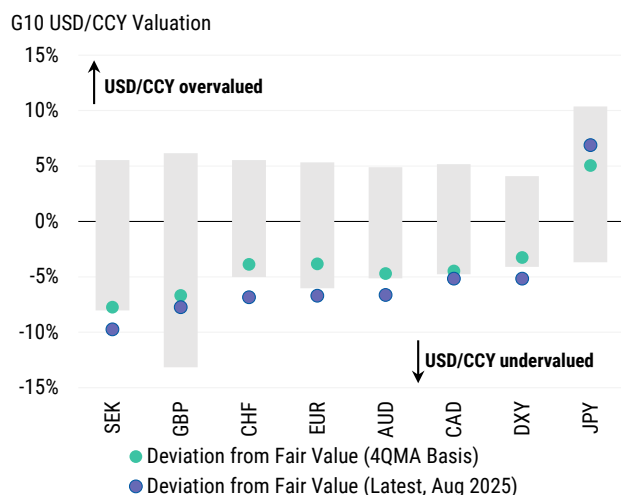


Source: Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

The picture is more nuanced across safe-haven currencies: SEK, GBP, CHF and EUR all screen rich to equilibrium, while JPY stands out as undervalued, underscoring its role as a global hedge currency. The DXY also seems undervalued by around 5% on a mark-to-market basis.

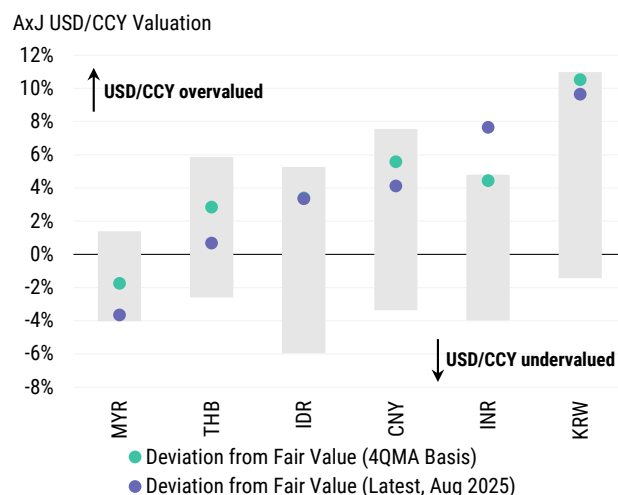
Among the commodity bloc, AUD and CAD both screen as moderately overvalued versus USD while BRL's overvaluation looks a bit extreme, most likely reflecting domestic risk premia layered on top of its commodity profile.

Exhibit 34: Most G10 currencies are overvalued versus USD



Source: Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

Exhibit 35: KRW and INR are most undervalued in Asia

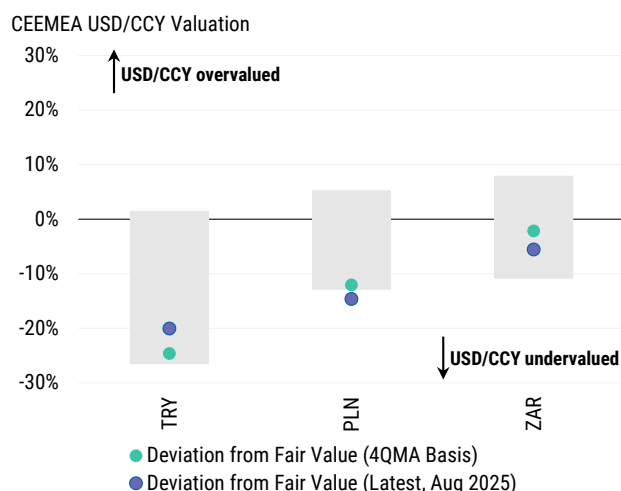


Source: Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates.

In Asia, KRW, INR and CNY stand out as the most undervalued versus USD: For USD/CNY, the downward mark-to-market adjustment underscores its policy-managed exchange

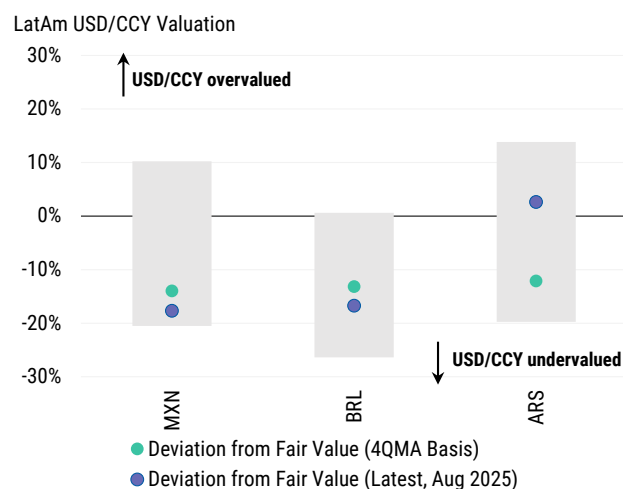
rate regime, while INR's stretched undervaluation against USD reflects ongoing trade and tariff tensions alongside equity outflows from foreign investors. Other Asian currencies, such as IDR and THB, appear more balanced, undervalued by 1-3%, whereas MYR screens as overvalued by about 4% against USD.

Exhibit 36: Valuations are a bit stretched in CEE...



Source: Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates

Exhibit 37: ...as well as in LatAm



Source: Morgan Stanley Research estimates; Note: Valuations and estimates are as of 2Q25 and presented on a 4QMA basis. Historical valuation range bar is capped at 2Q25 estimates

In summary, currency overvaluation seems concentrated in LatAm/CEE and parts of Europe, while most of Asia screens as undervalued versus USD, creating a barbelled valuation profile across currency groups.

Exhibit 38: G10 and EM FEER valuations

Region	REER			Currency Pair	USD/CCY		
	Latest REER (August 2025)	FEER based Estimate	Spot Deviation from Fair Value		Latest FX Cross (August 2025)	FEER based Estimate	Spot Deviation from Fair Value
TRY	98	77	27.9%	USD/TRY	41	51	-20.0%
MXN	128	105	22.2%	USD/MXN	18.7	22.7	-17.7%
BRL	108	90	20.0%	USD/BRL	5.4	6.5	-16.8%
PLN	126	111	13.7%	USD/PLN	3.6	4.3	-14.6%
EUR	105	97	8.5%	USD/SEK	9.5	10.5	-9.7%
SEK	98	91	8.2%	USD/GBP	0.74	0.80	-7.7%
GBP	112	105	6.7%	USD/CHF	0.80	0.86	-6.9%
MYR	101	96	5.6%	USD/EUR	0.86	0.92	-6.7%
CHF	104	99	4.4%	USD/AUD	1.53	1.64	-6.6%
CAD	97	93	4.1%	USD/ZAR	17.7	18.7	-5.6%
AUD	105	101	4.0%	USD/CAD	1.37	1.45	-5.2%
ZAR	104	101	2.5%	DX	97.8	103.1	-5.2%
THB	98	98	0.3%	USD/MYR	4.2	4.4	-3.7%
JPY	72	74	-2.5%	USD/THB	32.4	32.2	0.7%
ARS	126	131	-3.6%	USD/ARS	1344	1310	2.6%
IDR	95	100	-4.3%	USD/IDR	16490.0	15955.1	3.4%
USD	109	114	-4.6%	USD/CNY	7.13	6.85	4.1%
CNY	87	94	-7.4%	USD/JPY	147.1	137.6	6.9%
INR	96	105	-8.8%	USD/INR	88.21	81.94	7.7%
KRW	91	100	-9.6%	USD/KRW	1389.90	1267.50	9.7%

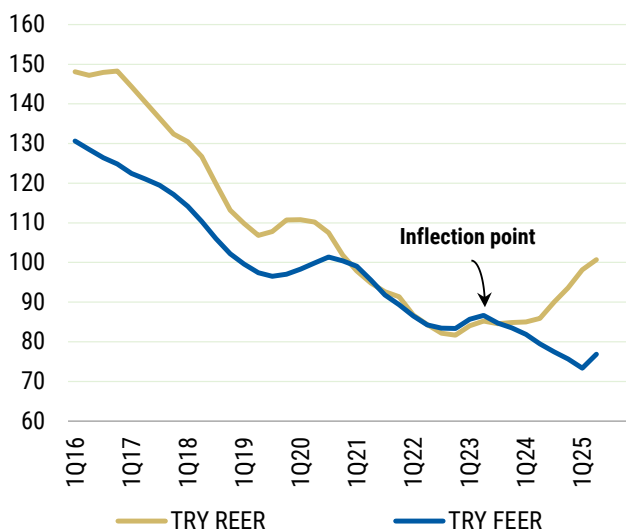
Source: Bloomberg, BIS, Morgan Stanley Research estimates

Why Do Some Currencies Diverge Away from FEER-Based Values?

Study 1: Turkey: A structural outlier

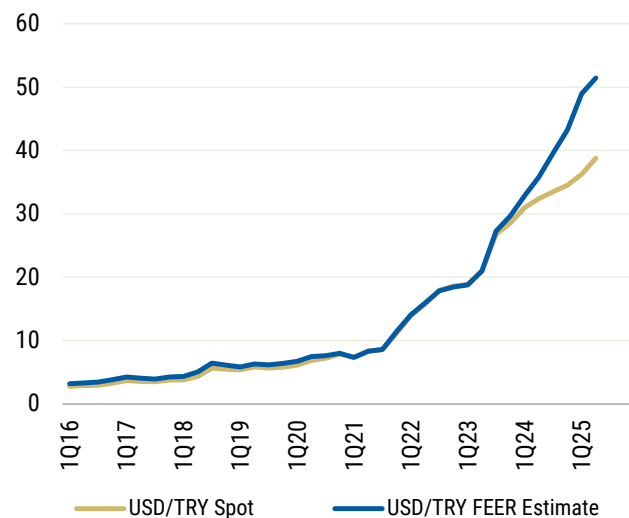
Our FEER model has shown a growing and persistent overvaluation of TRY post-2023, diverging markedly from our model-consistent estimates. This divergence appears to stem from a sequence of macroeconomic disruptions and policy responses that have undermined the model's underlying assumptions.

Exhibit 39: Turkey REER



Source: BIS, Morgan Stanley Research estimates

Exhibit 40: USD/TRY

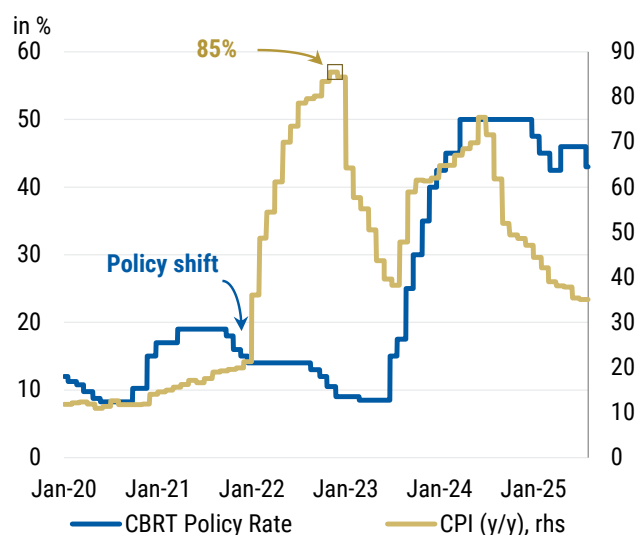


Source: Bloomberg, Morgan Stanley Research estimates

As seen from [Exhibit 40](#), since 2021, the TRY has undergone a multi-year depreciation episode, largely disconnected from medium-term current account fundamentals and price adjustments. During this period the currency was also influenced heavily by [FX-protected deposit schemes](#), [central bank interventions](#) and [heterodox monetary policy](#).

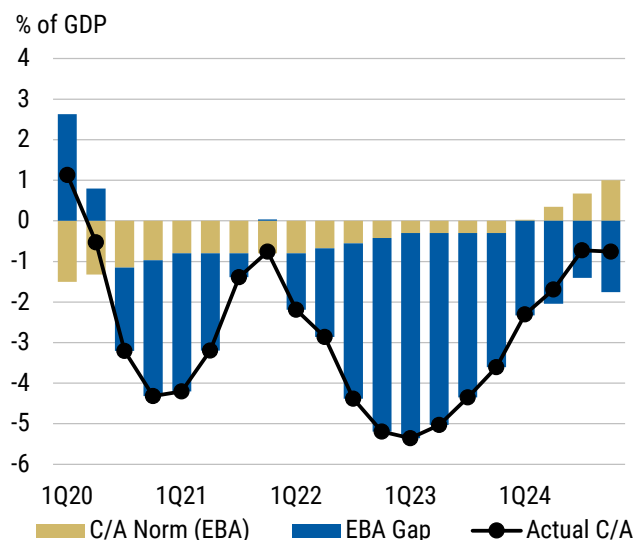
Excessive inflation and central bank policy: The CBT's decision to [cut policy rates aggressively](#) in late 2021, despite escalating inflation, pushed real interest rates into deeply negative territory. Inflation subsequently accelerated, surpassing 85%Y in 2022 and placing Turkey in an episode of excessive inflation ([Exhibit 41](#)). This severely distorted price signals, invalidating the assumption of relative price stability that underpins FEER modeling. Although the REER responded, its adjustment lagged the sharp nominal depreciation of TRY, resulting in significant valuation discrepancies between FEER estimates and the spot rate.

Exhibit 41: The CBT slashed rates even after inflation climbed over 20%Y



Source: Bloomberg, Morgan Stanley Research

Exhibit 42: Current account distortions



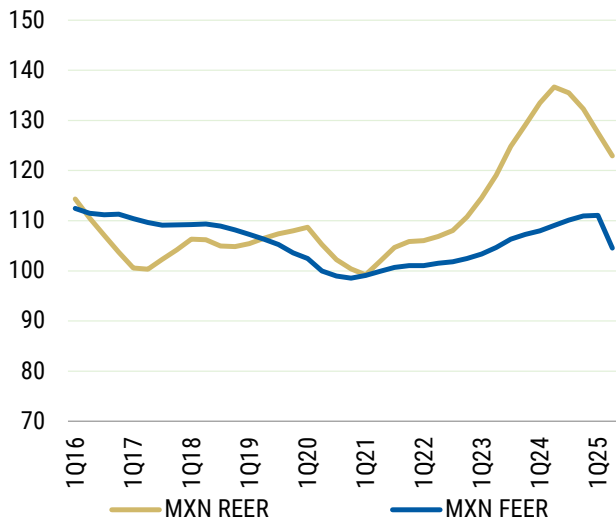
Source: IMF, Morgan Stanley Research

Structural current account distortion: While the FEER framework is anchored in a sustainable current account norm, Turkey's actual current account position deteriorated due to elevated post-Ukraine conflict [energy import costs](#), credit-fuelled domestic demand and [underperformance in tourism](#) and goods exports relative to import compression. This persistent external imbalance widened the gap between the actual current account and the EBA-consistent norm. Elevated country risk premiums embedded in TRY assets also contributed to excess depreciation not warranted by medium-term fundamentals.

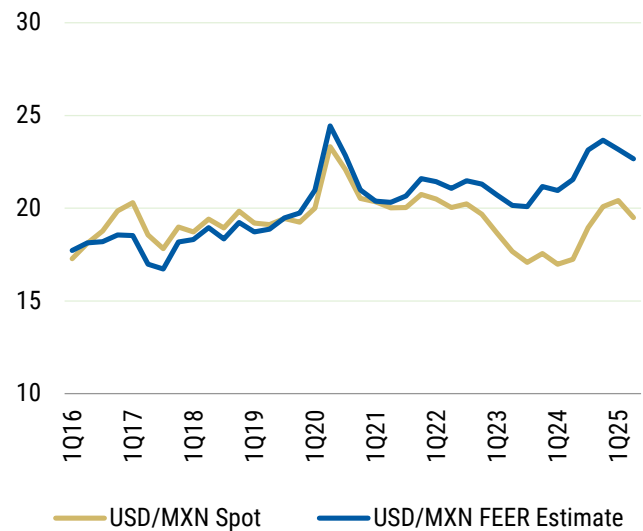
In essence, TRY is a structural outlier: The divergence in TRY's REER versus FEER post-2023 is primarily due to domestic policy regime shifts, capital account pressures and runaway inflation, none of which are fully internalized in a fundamentals-based FEER framework. Without augmenting the framework with these explicit measures, FEER valuations for TRY should be interpreted with particular caution.

Study 2: Mexico: FEER behind the curve

We observe a material divergence between MXN's REER and its FEER since 2020, showing signs of persistent overvaluation. In our view, this reflects a mix of cyclical capital flows and structural regime shifts that are not yet fully captured in our FEER framework ([Exhibit 43](#) and [Exhibit 44](#)).

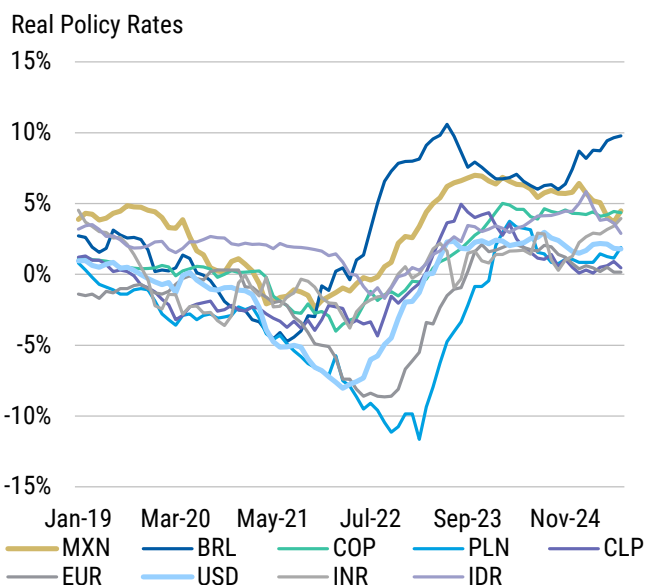
Exhibit 43: MXN REER

Source: BIS, Morgan Stanley Research estimates

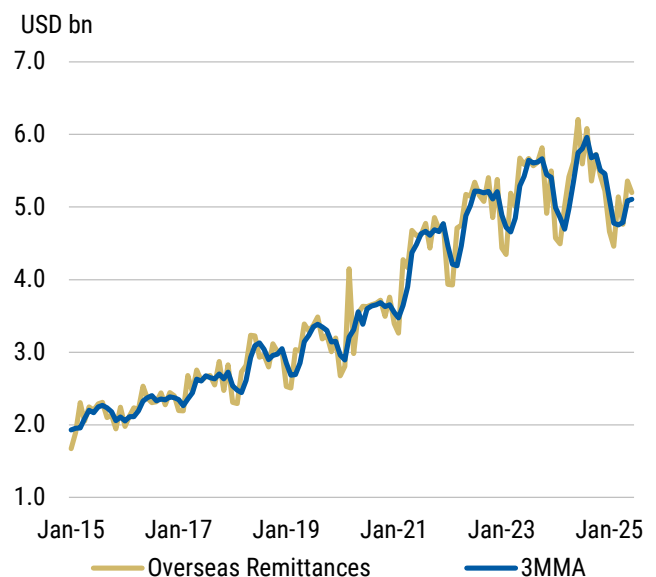
Exhibit 44: USD/MXN

Source: Bloomberg, Morgan Stanley Research estimates

The pull of yield and inflows: Banxico went on an early and aggressive tightening cycle, delivering one of the highest real policy rates in EM, drawing significant portfolio inflows to local bonds. Combined with Mexico's prudent fiscal profile, these inflows compressed some of the sovereign risk premia and supported MXN. Concurrently, record remittances and a swift tourism rebound provided a steady stream of FX inflows, amplifying the REER well beyond FEER-implied levels.

Exhibit 45: Post-2020, Mexico had one of the highest real policy rates

Source: Bloomberg, Morgan Stanley Research

Exhibit 46: Steady remittance inflows provide a constant flow of foreign exchange to Mexico

Source: Banxico, Morgan Stanley Research

Structural re-rating via nearshoring: Mexico experienced a structural re-rating of its external accounts, driven by the USMCA framework (agreed in [October 2018](#), entered in

2020) and [global supply-chain realignment](#). The nearshoring trend catalyzed FDI inflows, which not only validate current valuations but also imply a stronger future current account trajectory than EBA-based sustainable norms assume.

Exhibit 47: Foreign holdings of MBonos versus real policy rates

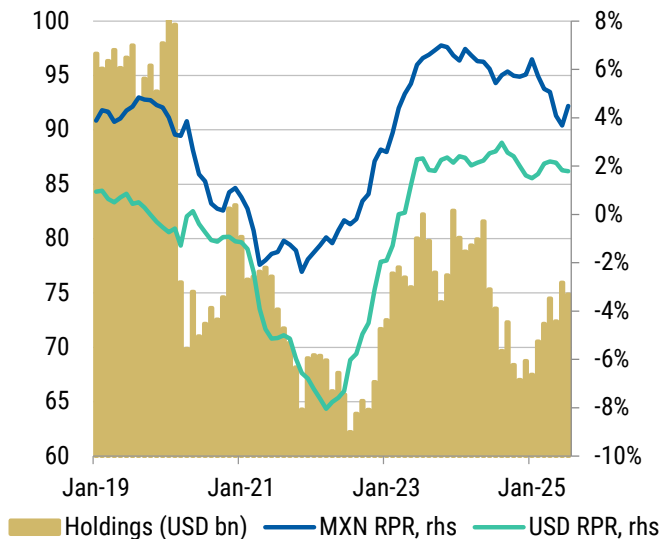
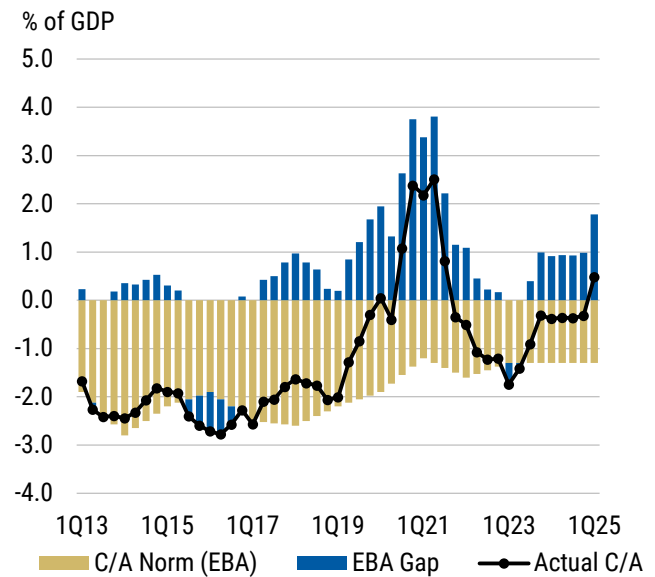


Exhibit 48: Current Account improvements post-USMCA



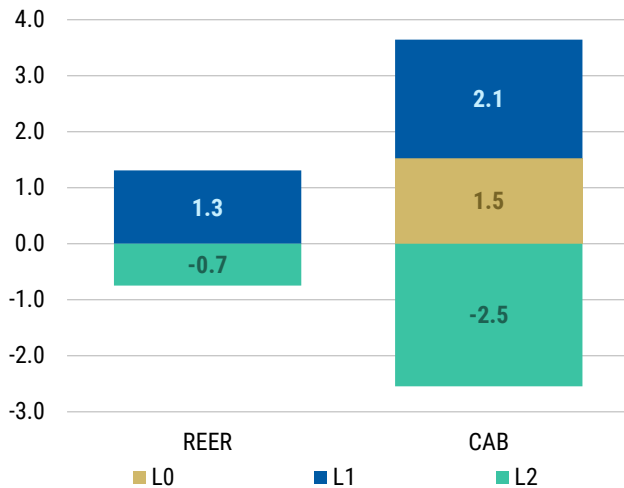
Study 3: What's driving the dollar?

Our ARDL results for the USD REER point to an AR(2) process with alternating sign ([Exhibit 49](#)). The first lag enters positive, while the second lag enters negative. This structure suggests a stationary but oscillatory adjustment path. In other words, rather than drifting smoothly back to equilibrium, shocks to the dollar tend to carry forward with momentum before corrective forces emerge (with a lag). This produces oscillatory swings in which the currency initially overshoots its fair value and only gradually converges back towards equilibrium.

Historically such behavior looks consistent with long swings witnessed in USD: For example, in the late 1990s and early 2000s, the tech boom, foreign investment and safe-haven demand propelled USD higher, before a weakening phase realigned it with underlying fundamentals ([Exhibit 50](#)).

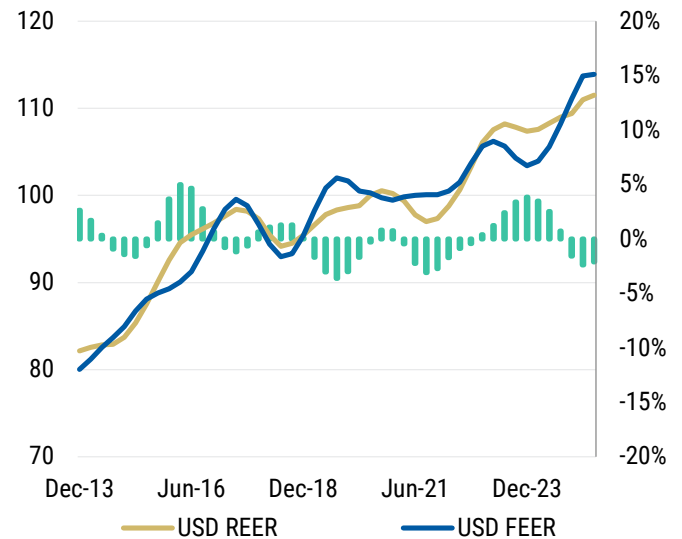
More recently, the 2014-17 dollar rally reflected monetary divergence, safe-haven flows and commodity shocks, underscoring the same pattern: The rally was fueled primarily by monetary policy divergence, as the Fed ended QE and prepared to hike rates while the ECB and BoJ eased aggressively. This was reinforced by safe-haven flows, terms of trade shifts and EM stresses (the period of sharp CNY depreciation, for example), as well as a sharp downturn in commodity exporters' currencies. Together, these forces drove a rapid appreciation of USD beyond equilibrium levels.

Exhibit 49: AR(2) and DL(2) process suggests a stationary but oscillatory adjustment path for USD REER



Source: Morgan Stanley Research

Exhibit 50: USD has a tendency to move in long swings around its FEER



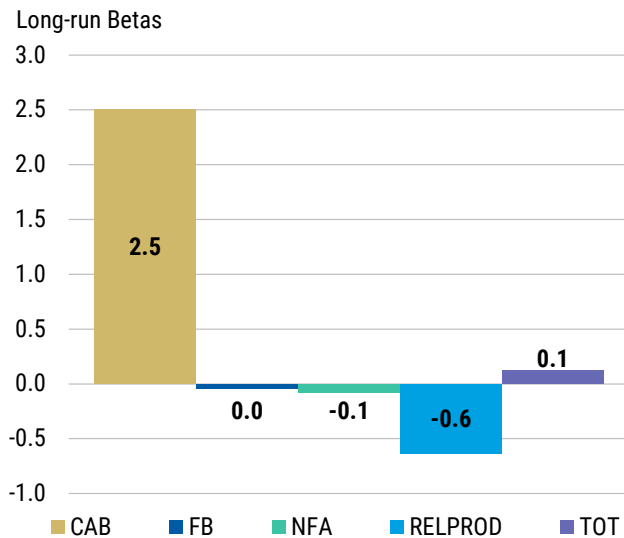
Source: BIS, Morgan Stanley Research estimates

By 2020-21, the dollar had slipped below fair value: Following the post-2014 rally, USD began to correct, and by 2017-18 the REER was trading broadly in line with fair value. This was followed by a period of undervaluation, as narrowing US rate differentials, escalating trade tensions and eventually the extraordinary policy response to Covid weighed on the currency. The combination of large fiscal stimulus and loose monetary policy briefly pushed USD below its estimated FEER level.

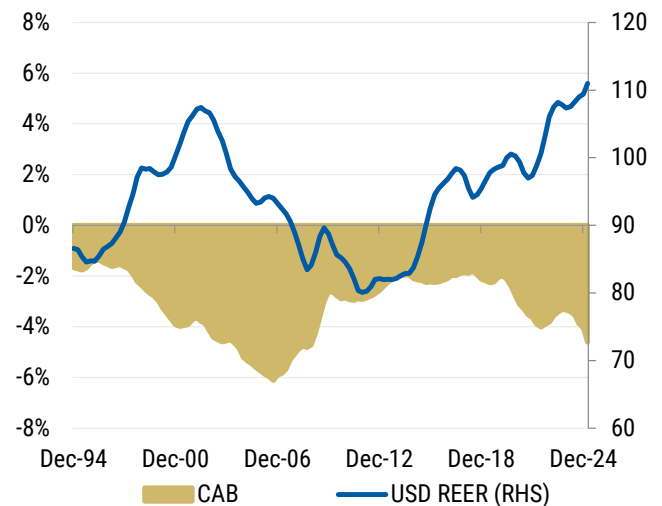
This undervaluation phase gave way to a sharp reversal, with USD swinging back into overvalued territory by 2023-24. The move was driven mainly by aggressive Fed tightening, which pushed US real yields sharply higher, alongside global energy shocks and geopolitical tensions that reinforced USD's safe-haven demand. These forces propelled the USD REER well above its FEER-implied level, underscoring the currency's tendency to overshoot fundamentals during periods of policy divergence and elevated risk aversion.

The current account as the core long-run driver

Within the model, the long-run betas point to the current account balance (CAB) as the central anchor for USD's fair value ([Exhibit 51](#)). In contrast, fiscal stance, terms of trade and net foreign assets register only a marginal influence, while relative productivity plays a modest and less intuitive role. This outcome is consistent with USD's unique reserve currency status, where capital account dynamics and global demand for USD assets often overshadow the conventional trade and productivity channels.

Exhibit 51: USD REER drivers

Source: Morgan Stanley Research

Exhibit 52: USD REER versus CAB

Source: BIS, IMF, Morgan Stanley Research estimates

Short-run dynamics of the CAB indicate a DL(2) process with a distinct overshooting pattern: The impact coefficient (LO) is positive and reinforced at the first lag (L1), before correcting sharply at the second lag (L2). This suggests that while CAB improvements tend to strengthen the USD REER, the adjustment is non-linear and cyclical, often front-loaded before partial reversal ([Exhibit 52](#)).

The historical interplay between the CAB and USD REER also points to a cyclical adjustment process rather than a linear one: Periods of USD strength, such as the late 1990s and post-2014 rallies, were associated with widening current account deficits, underscoring how capital inflows and safe-haven demand allow the US to run larger deficits while the currency appreciates. Conversely, phases of dollar weakness, most notably in the early 2000s and following the global financial crisis, align with a partial narrowing of the deficit.

The 2020-21 Covid shock reinforced this pattern: The REER softened briefly even as extraordinary fiscal stimulus and ultra-loose Fed policy widened the external deficits. The subsequent Fed tightening cycle and heightened risk-off sentiment then drove USD sharply higher, despite the CAB slipping further into deficit.

The dollar's paradox: Why fair value rose as the current account deficit widened?

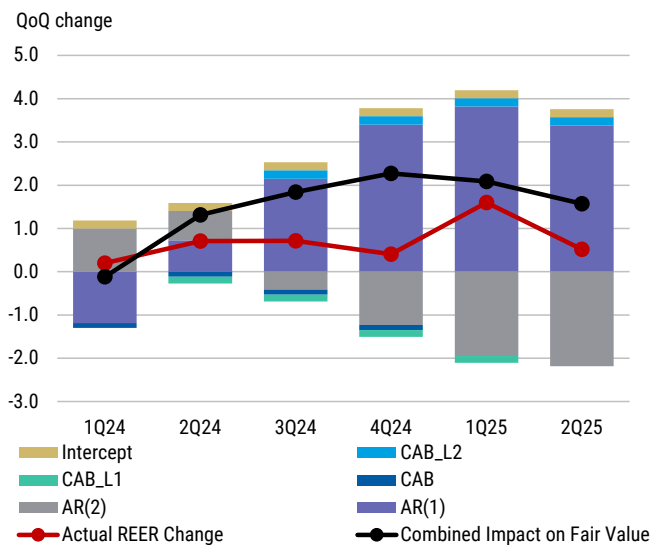
USD's behavior in early 2024 underscores a striking paradox: our model indicates that its fair value has been rising even as the US current account deficit has widened. While this may appear counter-intuitive at first glance, the explanation lies in the distinction between the short-term forces that drive the dollar and the long-term fundamentals that ultimately anchor it. As shown in [Exhibit 53](#), we disentangle these dynamics to illustrate why fair value moved higher despite a deteriorating external balance.

In 1Q24 and thereafter, USD's REER dynamics were driven largely by persistence, with the autoregressive terms indicating that most quarter-to-quarter movements were path

dependent. Put simply, once USD strengthened on the back of Fed tightening (via higher US Treasury yields), global risk-off sentiment and sustained safe-haven demand, that momentum carried forward into subsequent quarters.

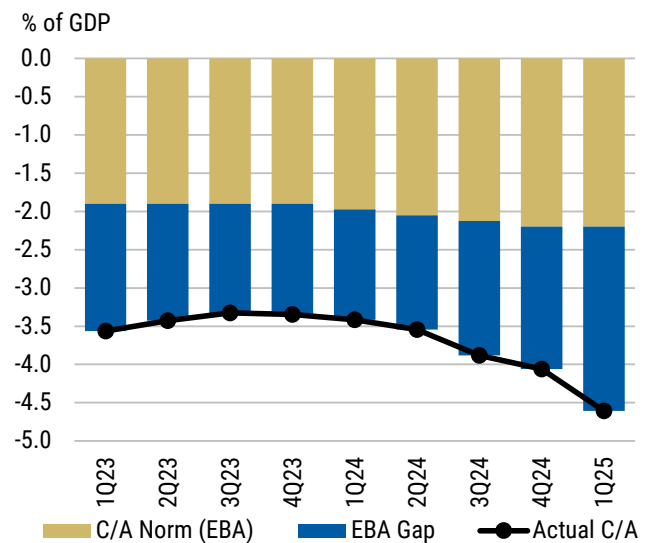
From the model's perspective, this self-reinforcing mechanism, with the first lag entering positively and the second lag negatively, emerges as the dominant driver of short-term fair value changes, while the current account balance appears to have only a modest impact in the near term.

Exhibit 53: USD fair value contribution factors



Source: BIS, Morgan Stanley Research

Exhibit 54: US current account balance and EBA-CA norms



Source: IMF, Morgan Stanley Research

At the same time, our model highlights the current account as the fundamental anchor for the dollar's equilibrium ([Exhibit 51](#)). The long-run beta of the CAB is estimated at +2.5, implying that a permanent 1-unit improvement in the US current account balance raises the equilibrium USD REER by nearly 2.5 units. This reinforces the point that while the CAB's influence looks modest in the quarter-to-quarter dynamics, its cumulative effect is substantial.

The dollar cannot drift indefinitely on momentum alone; over time it must converge towards levels consistent with the external balance. Put differently, the CAB determines the 'destination', even if the REER's own persistence dominates the 'journey'.

In summary, the rise in USD's fair value alongside a widening current account deficit is not a contradiction, but rather a reflection of the model's structure. Valuation signals for USD remain strong, though the timing matters. In the near term, inertia and capital flows continue to be major drivers, while over the longer horizon the current account serves as the anchor that ultimately determines where the dollar must settle.

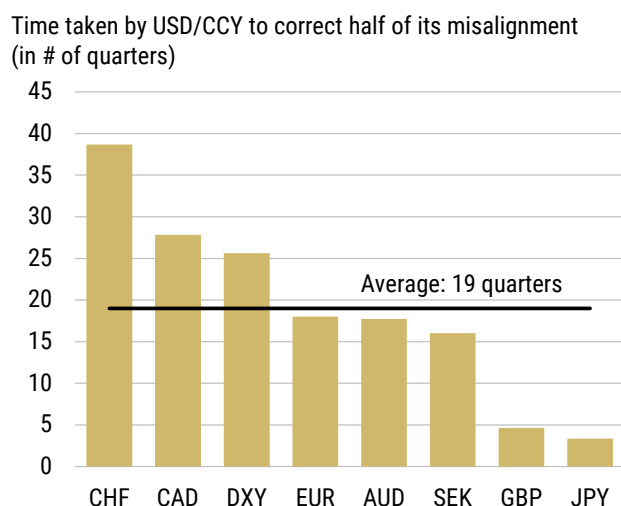
For investors, the trade-off is clear: ride the momentum in the short run, but remain mindful of reversion towards fundamentals over the medium-to-long term.

Convergence to Fair Value

To assess the speed at which exchange rates converge to their estimated fair value, we examine the persistence of misalignments between spot USD/CCY rates and their FEER-implied levels. We define the misalignment as the log deviation of the spot rate from its FEER and estimate an autoregressive process of order one (AR(1)) on this series to capture the dynamic adjustment process.

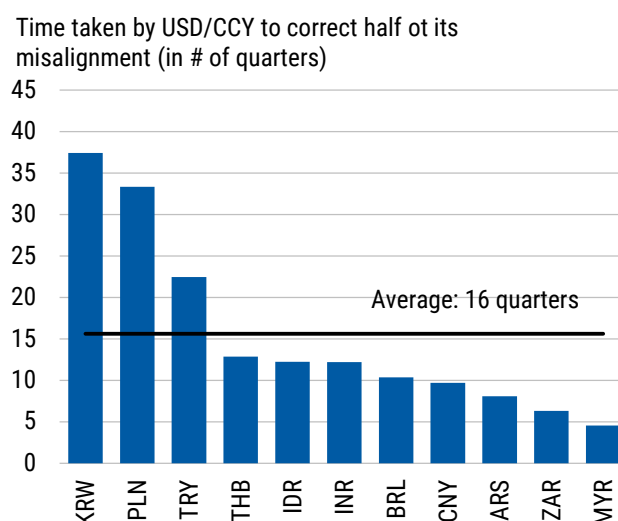
The autoregressive beta provides a measure of how quickly deviations from fair value tend to correct over time. From this, we calculate the half-life of misalignment (i.e., the time it takes for half of the initial deviation to be eliminated), which serves a good proxy of convergence speed.

Exhibit 55: In the G10, CHF and CAD take the longest to converge to their fair value



Source: Morgan Stanley Research

Exhibit 56: In EM, most currencies converge relatively quickly to fair value



Source: Morgan Stanley Research

Currencies take approximately 16-19 quarters to correct half of their misalignments:

Within the G10, CHF and CAD exhibit the slowest pace of convergence, followed by the DXY. In contrast, most other currencies adjust more rapidly, with an average half-life of around 12 quarters. Notably, GBP and JPY show the fastest correction speeds, at five and three quarters, respectively.

In EMs, KRW, PLN and TRY display the slowest adjustment. Asian currencies such as THB, IDR and INR follow while ZAR and MYR stand out for their relatively faster mean-reversion.

The distinction between G10 and EM currencies' convergence profile likely reflects differences in their market depths and policy frameworks.

G10 currencies typically serve as reserve or funding currencies, supported by deeper, more liquid financial markets and more entrenched investor positioning. These structural characteristics can allow valuation misalignments to persist for extended periods.

In contrast, EM currencies are generally more flow driven and policy sensitive. The FX reaction function is often an explicit part of EM central bank toolkits, making FX dynamics more reactive. As a result, valuation gaps in EM currencies can correct relatively more swiftly, either through direct central bank intervention or broader policy adjustments.

Limitations of the FEER Model

Like every valuation framework, FEER comes with important caveats that must be kept in mind:

(1) Structural clarity, not short-term precision: The FEER framework is designed to highlight long-term equilibrium misalignments, but it does not capture near-term market dynamics. Cyclical drivers such as capital flows, risk sentiment and policy surprises can dominate currency moves for extended periods, meaning that convergence to 'fair value' is often slow and uneven. FEER should therefore be viewed as a strategic anchor, not a tactical timing tool.

(2) Reliance on normative benchmarks: Our FEER implementation uses current account norms from the IMF's EBA as proxies for sustainable balances. While widely accepted, these benchmarks are subject to methodological revisions and may not fully capture country-specific structural shifts, policy choices or political constraints. This injects a layer of uncertainty into the model's estimates.

For example, the IMF EBA assessments include "institutional quality" as one of several variables to explain what an economy's sustainable current account position might be, finding that "a country at the 75th percentile of the institutional quality distribution would have, all else equal, a 0.5 percentage points of GDP lower current account balance".

(3) Multilateral versus bilateral disconnects: FEER is built on the REER, which reflects trade-weighted competitiveness across many partners. As a result, signals from FEER may diverge from bilateral exchange rates, particularly when USD or other major trading partner currencies are themselves misaligned.

(4) Sensitivity to model specification: Estimated misalignments can vary meaningfully depending on the choice of variables, data sources or econometric specification. Differences in how productivity, fiscal stance or net foreign assets are measured can shift equilibrium estimates, underlining the importance of interpreting results with caution.

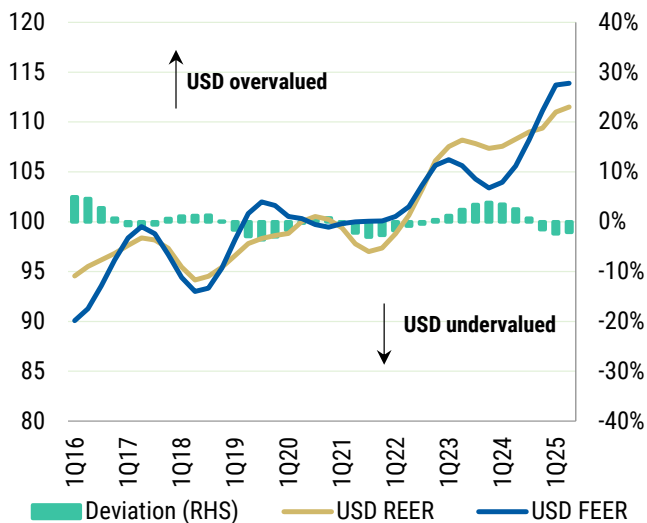
(5) Limited treatment of structural change: FEER frameworks assume stable long-run relationships between fundamentals and the REER. In practice, structural changes such as supply-chain realignments, demographic shifts or transitions in the global financial architecture can alter these relationships over time, making past coefficients a noisy guide to the future.

Looking ahead, future refinements of the FEER framework could focus on bridging the gap between long-run equilibrium and shorter-term dynamics. Incorporating capital flow variables and risk sentiment indicators would help to capture financial drivers that often dominate currency moves in practice.

Appendix 1: REER Valuation Charts

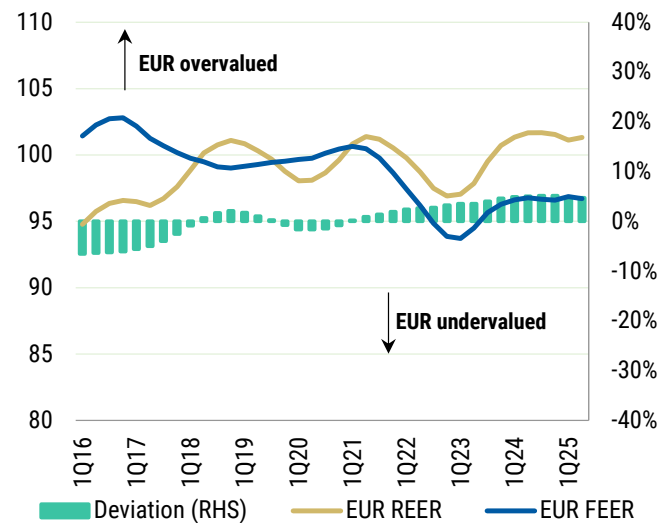
G10

Exhibit 57: USD



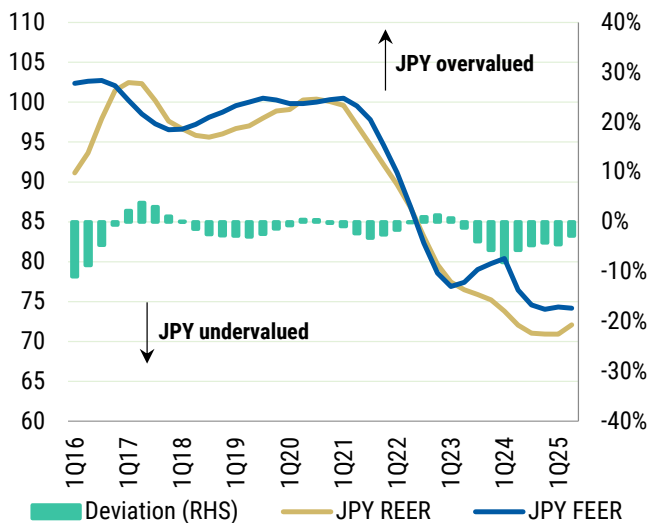
Source: BIS, Morgan Stanley Research estimates

Exhibit 58: EUR



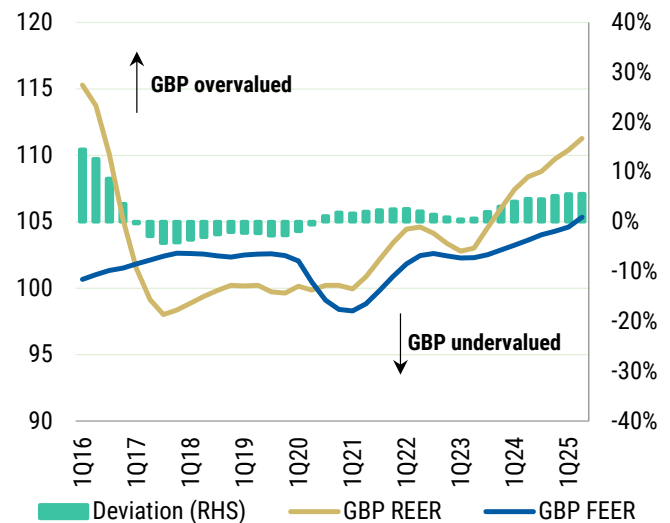
Source: BIS, Morgan Stanley Research estimates

Exhibit 59: JPY



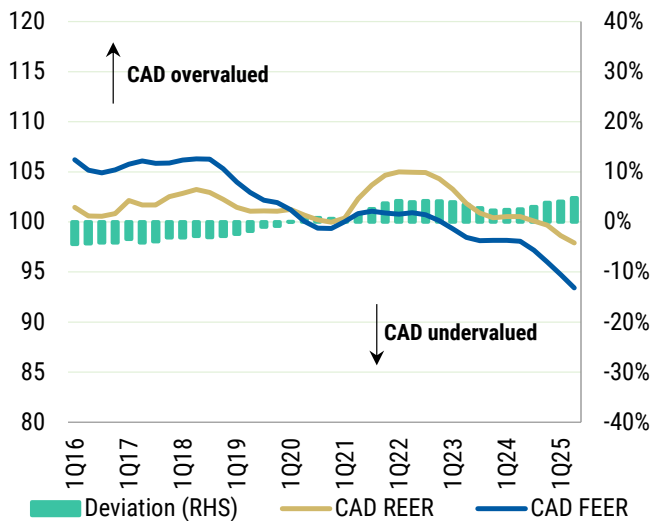
Source: BIS, Morgan Stanley Research estimates

Exhibit 60: GBP



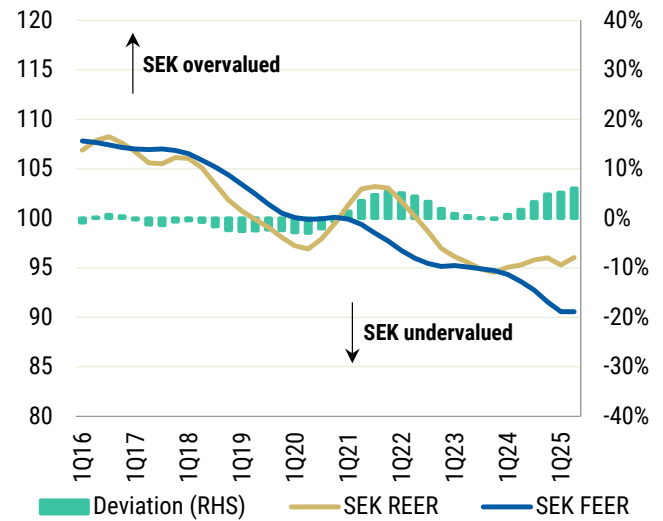
Source: BIS, Morgan Stanley Research estimates

Exhibit 61: CAD



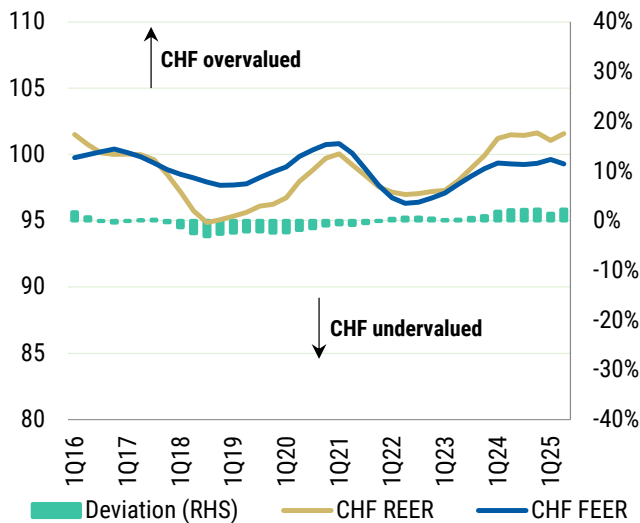
Source: BIS, Morgan Stanley Research estimates

Exhibit 62: SEK



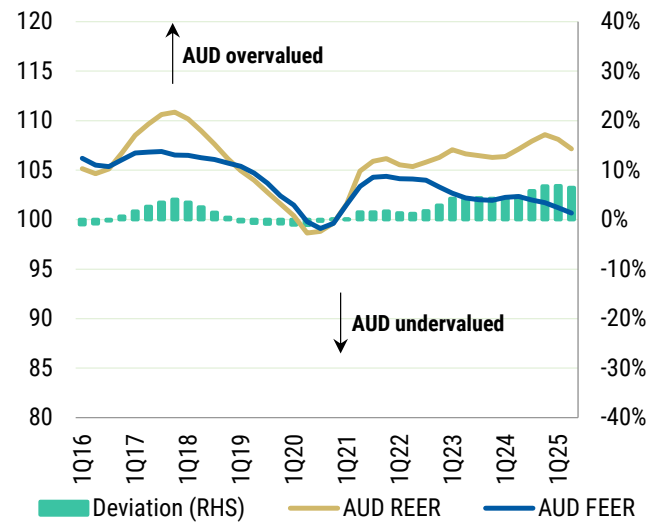
Source: BIS, Morgan Stanley Research estimates

Exhibit 63: CHF



Source: BIS, Morgan Stanley Research estimates

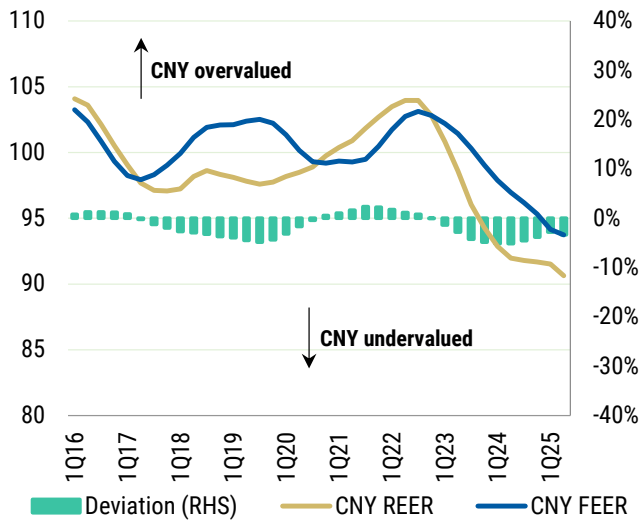
Exhibit 64: AUD



Source: BIS, Morgan Stanley Research estimates

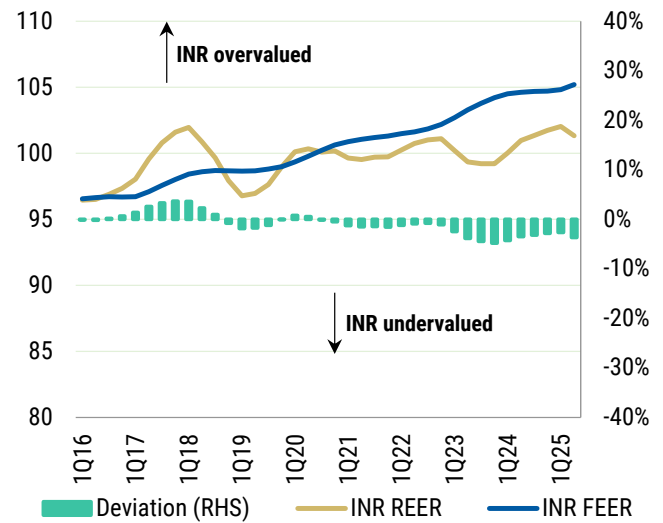
Axj

Exhibit 65: CNY



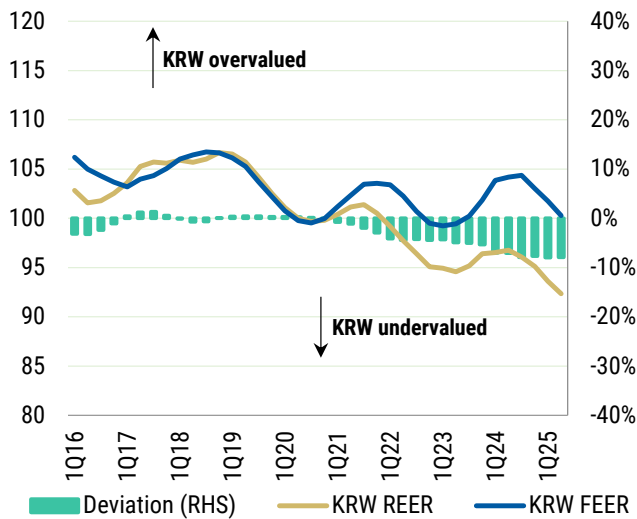
Source: BIS, Morgan Stanley Research estimates

Exhibit 66: INR



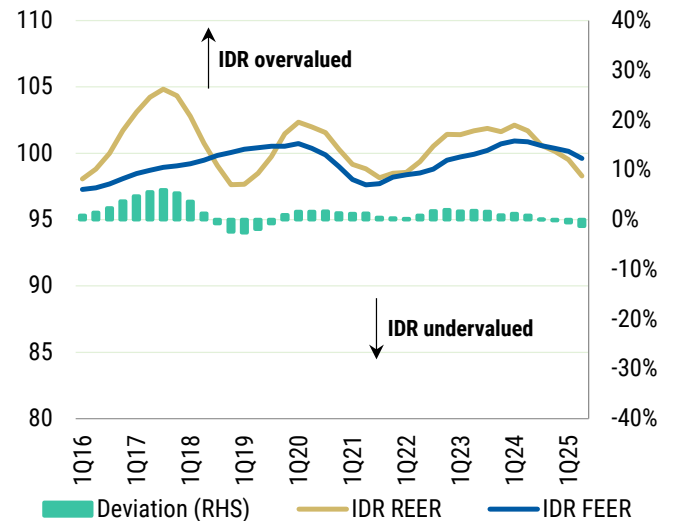
Source: BIS, Morgan Stanley Research estimates

Exhibit 67: KRW



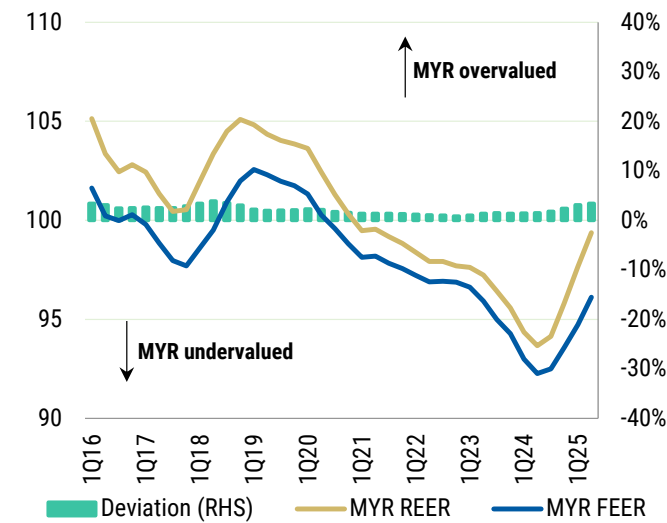
Source: BIS, Morgan Stanley Research estimates

Exhibit 68: IDR



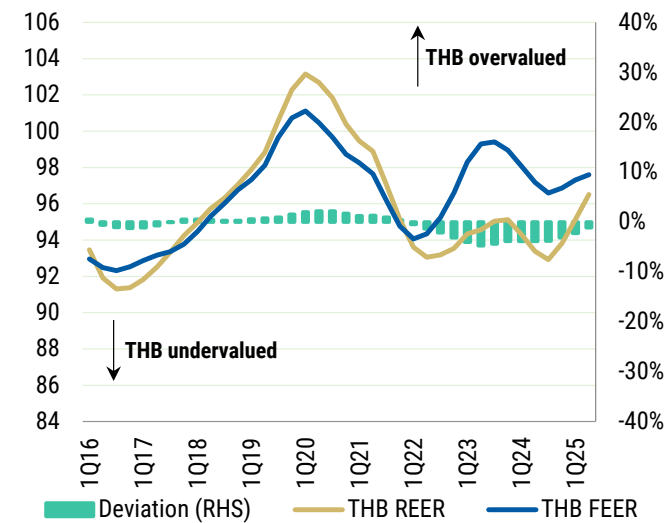
Source: BIS, Morgan Stanley Research estimates

Exhibit 69: MYR



Source: BIS, Morgan Stanley Research estimates

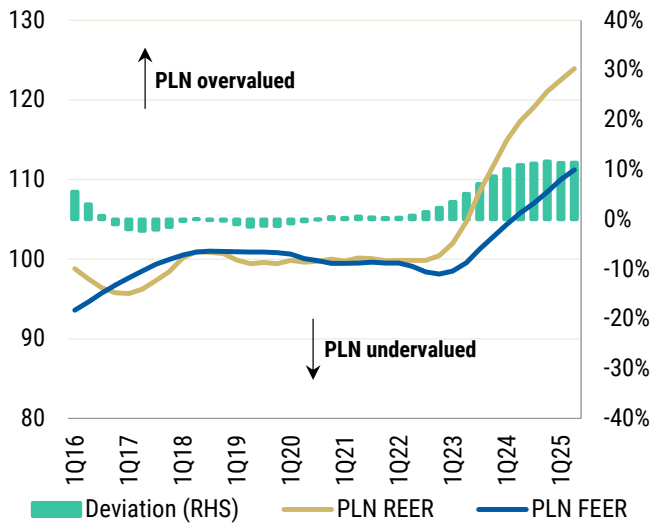
Exhibit 70: THB



Source: BIS, Morgan Stanley Research estimates

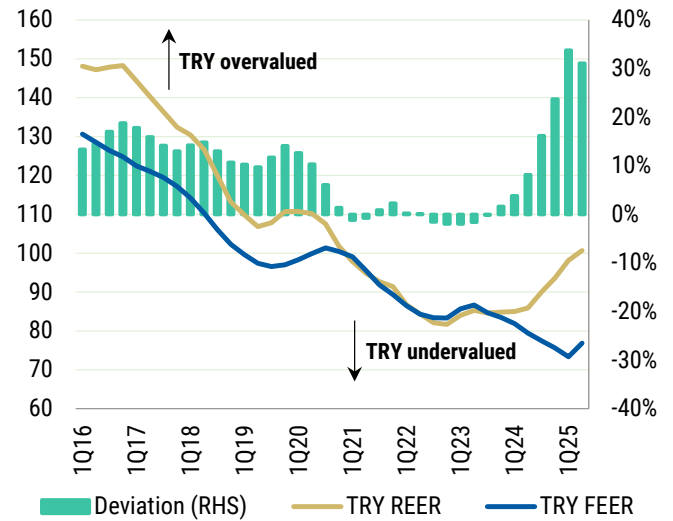
CEEMEA

Exhibit 71: PLN



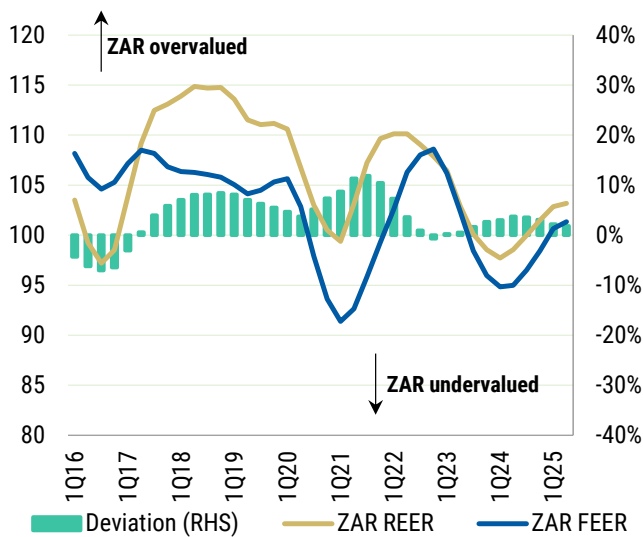
Source: BIS, Morgan Stanley Research estimates

Exhibit 72: TRY



Source: BIS, Morgan Stanley Research estimates

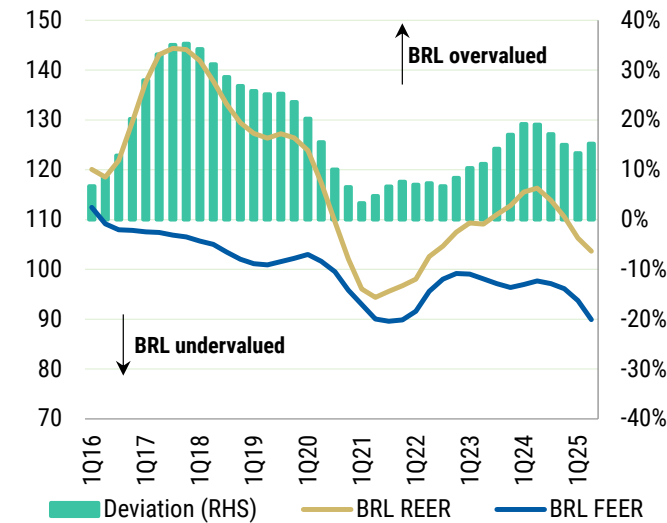
Exhibit 73: ZAR



Source: BIS, Morgan Stanley Research estimates

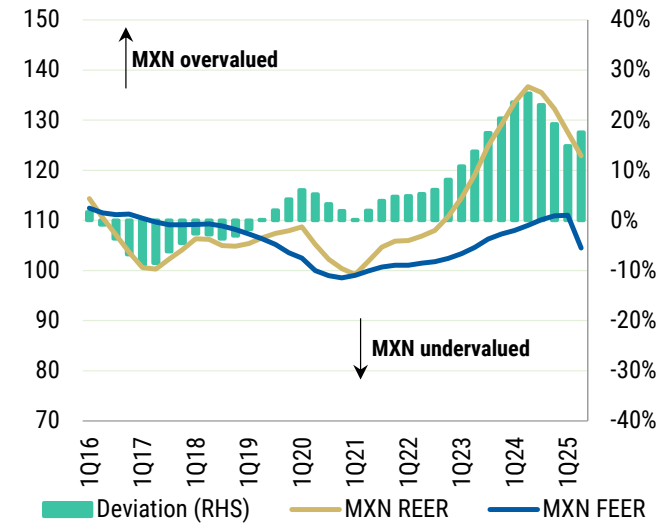
LatAm

Exhibit 74: BRL



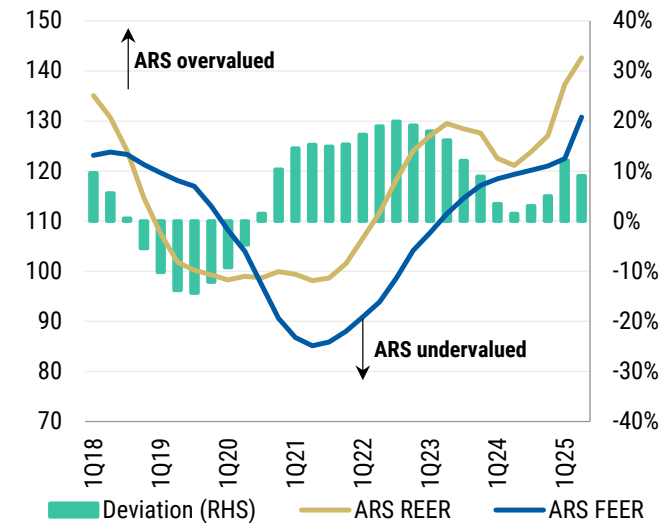
Source: BIS, Morgan Stanley Research estimates

Exhibit 75: MXN



Source: BIS, Morgan Stanley Research estimates

Exhibit 76: ARS

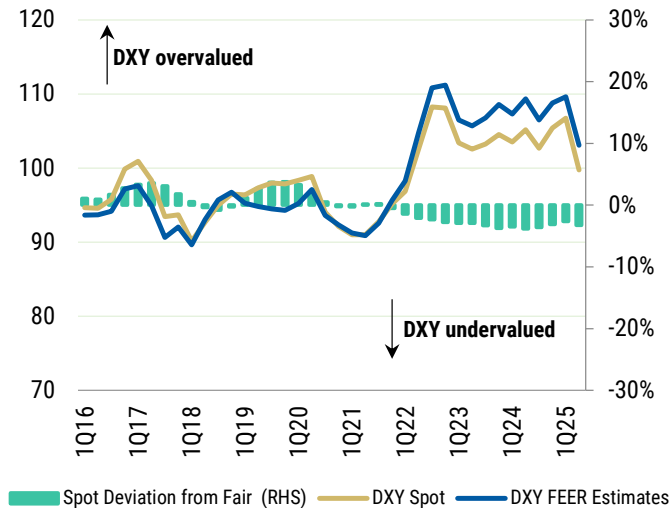


Source: BIS, Morgan Stanley Research estimates

Appendix 2: USD/CCY Valuation Charts

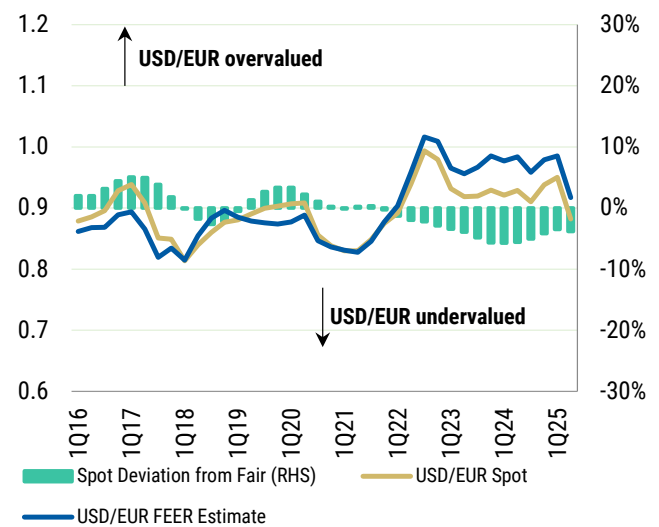
G10

Exhibit 77: DXY



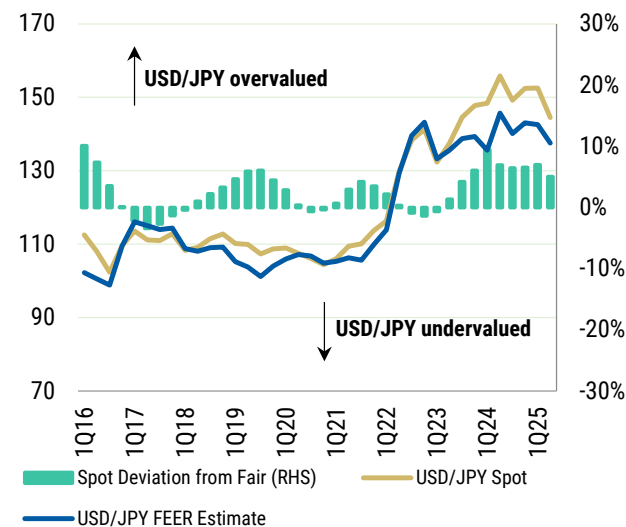
Source: Morgan Stanley Research estimates

Exhibit 78: USD/EUR



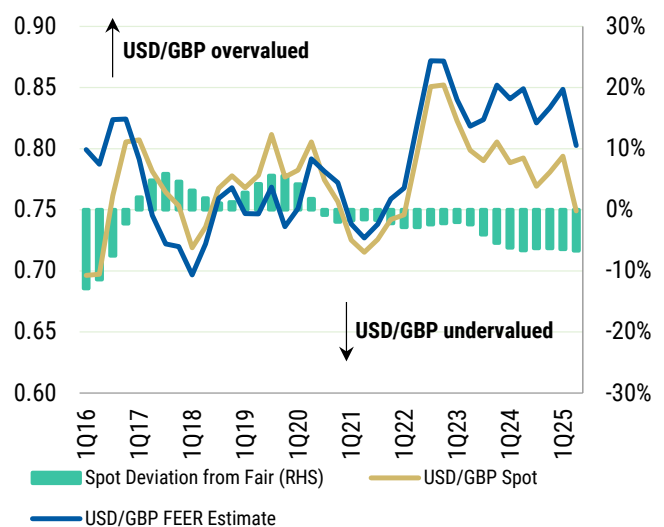
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 79: USD/JPY



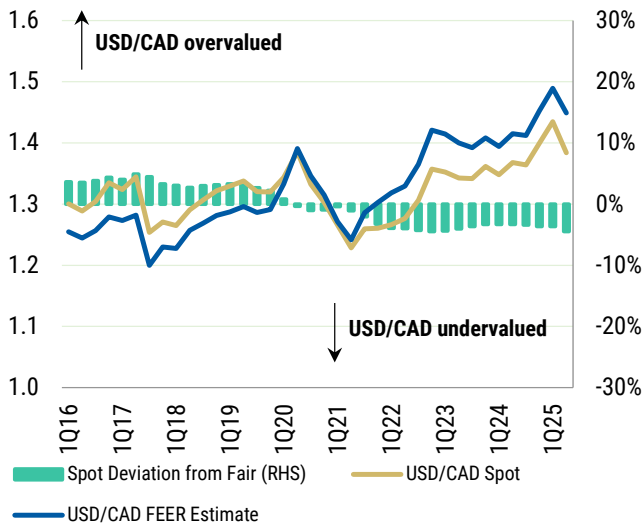
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 80: USD/GBP



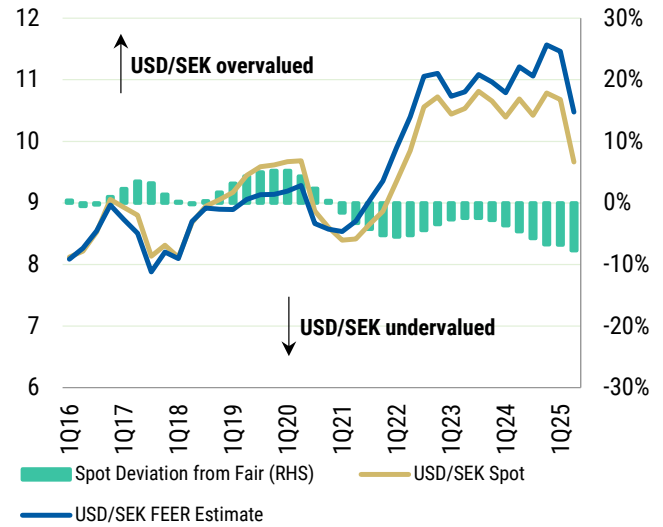
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 81: USD/CAD



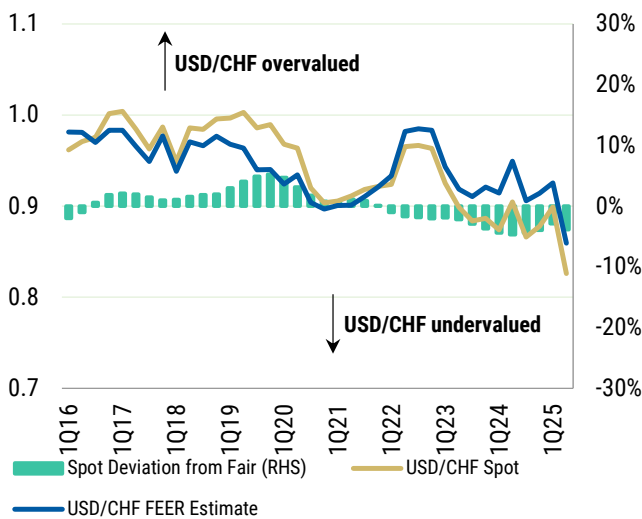
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 82: USD/SEK



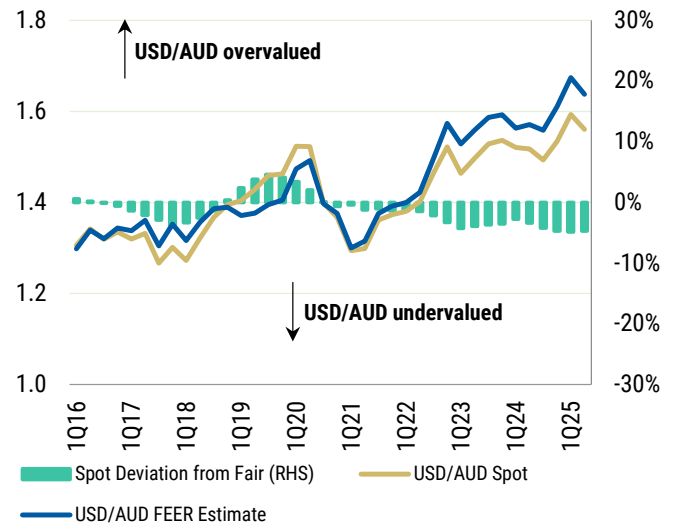
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 83: USD/CHF



Source: Bloomberg, Morgan Stanley Research estimates

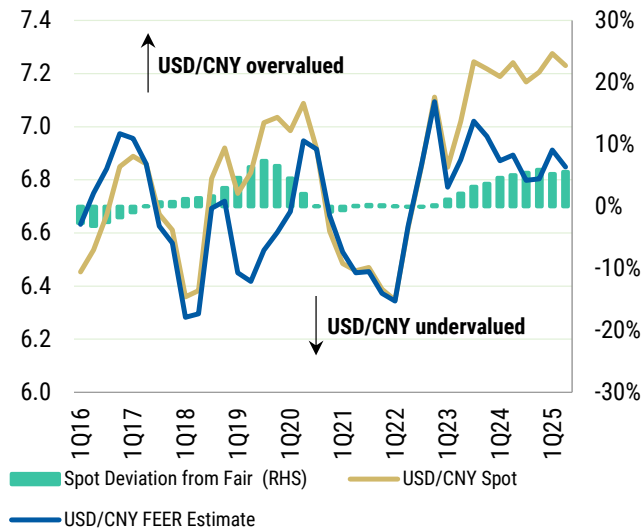
Exhibit 84: USD/AUD



Source: Bloomberg, Morgan Stanley Research estimates

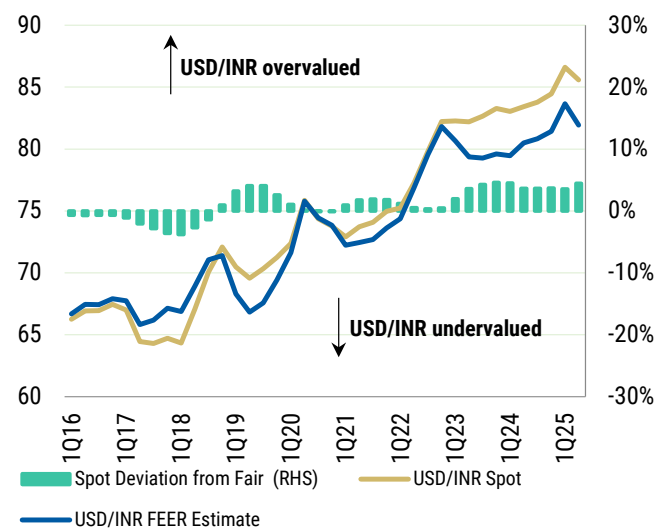
AxJ

Exhibit 85: USD/CNY



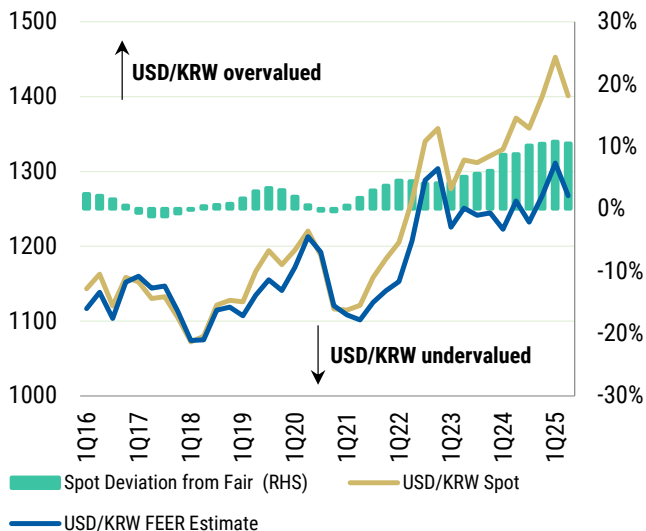
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 86: USD/INR



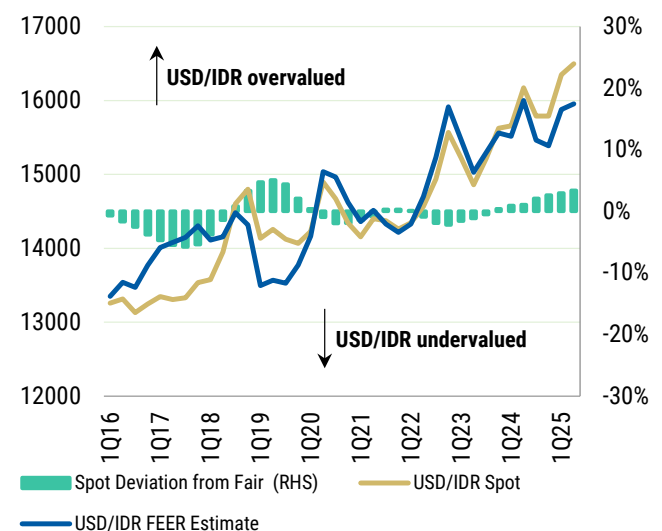
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 87: USD/KRW



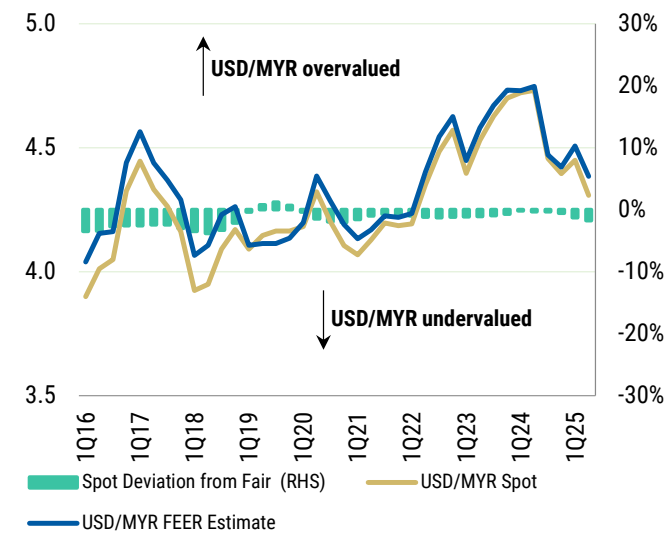
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 88: USD/IDR



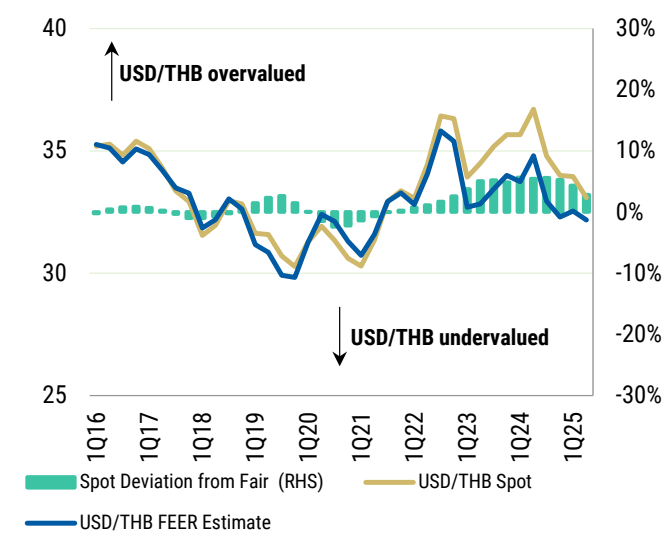
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 89: USD/MYR



Source: Bloomberg, Morgan Stanley Research estimates

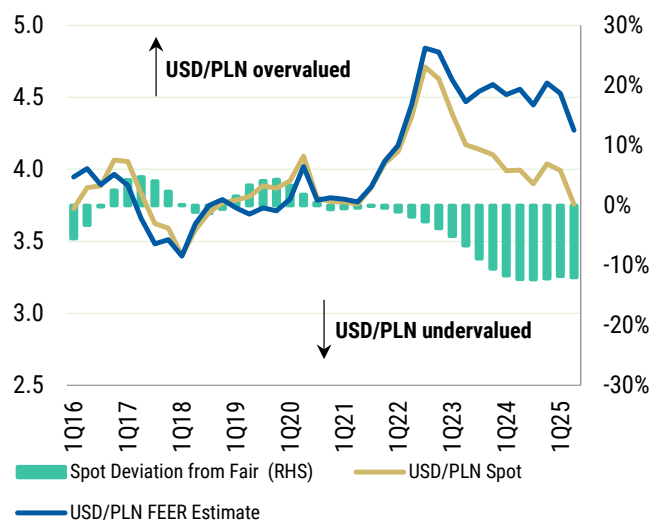
Exhibit 90: USD/THB



Source: Bloomberg, Morgan Stanley Research estimates

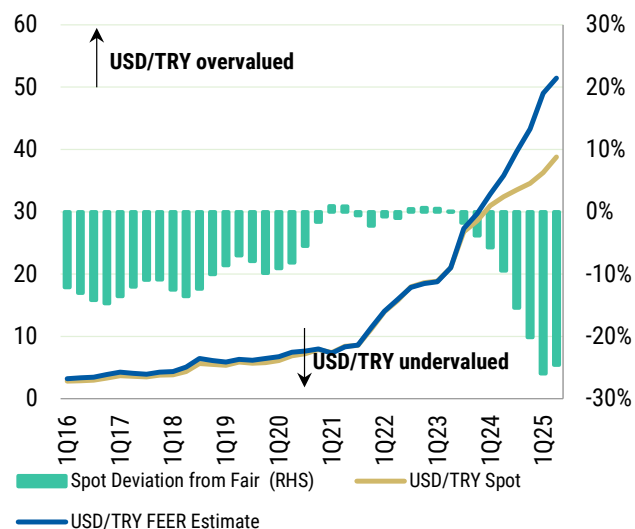
CEEMEA

Exhibit 91: USD/PLN



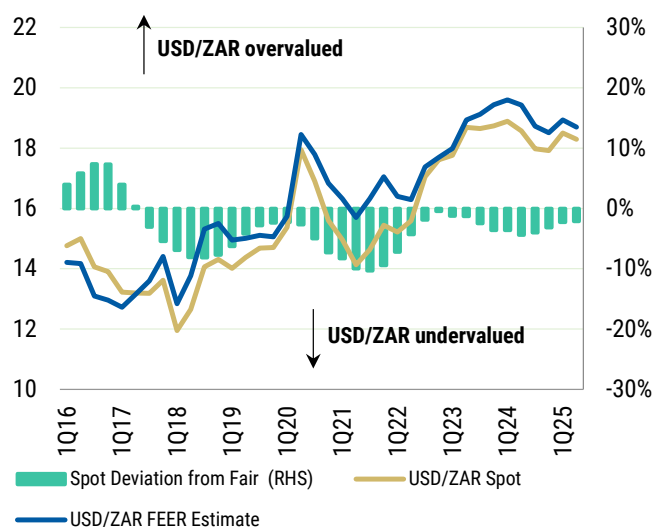
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 92: USD/TRY



Source: Bloomberg, Morgan Stanley Research estimates

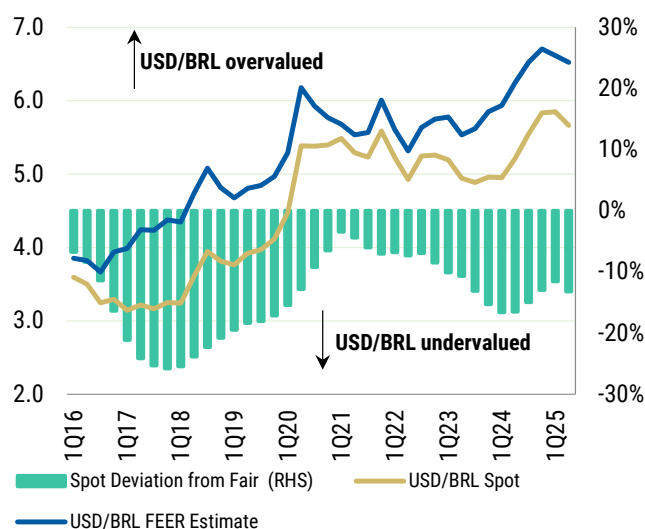
Exhibit 93: USD/ZAR



Source: Bloomberg, Morgan Stanley Research estimates

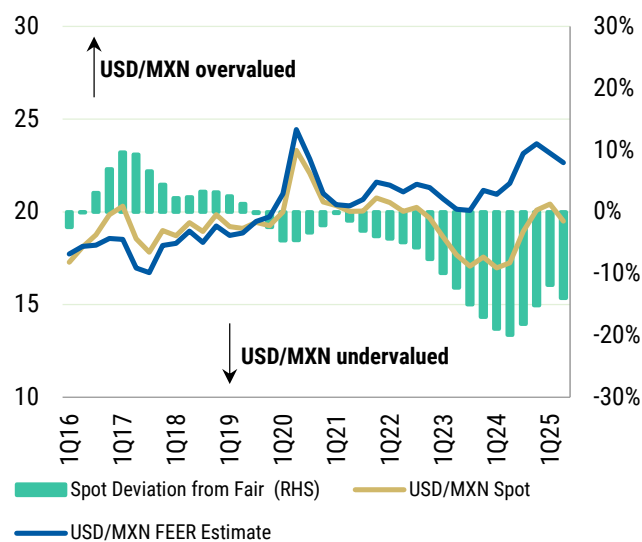
LatAm

Exhibit 94: USD/BRL



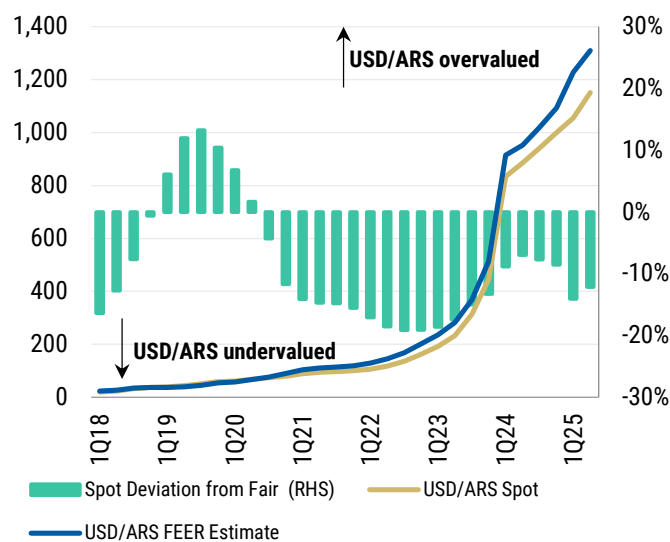
Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 95: USD/MXN



Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 96: USD/ARS



Source: Bloomberg, Morgan Stanley Research estimates

Appendix 3: Regression Betas

Exhibit 97: ARDL regression betas

	Real Effective Exchange Rate			Current Account Balance				Fiscal Balance				Net Foreign Asset Position				Relative Productivity				Terms of Trade			
	L1	L2	L3	L0	L1	L2	L3	L0	L1	L2	L3	L0	L1	L2	L3	L0	L1	L2	L3	L0	L1	L2	L3
USD	1.31	-0.75		1.53	2.12	-2.54		-0.02				0.00	-0.01	-0.01	-0.02	-0.02	-0.27			0.05			
EUR	1.33	-0.58		0.08				-0.07				-0.01				0.19				0.13			
JPY	1.39	-0.63		-1.05	1.15			-0.03				-0.05				-0.53				0.05			
GBP	1.45	-0.66		0.33				0.08				0.00				0.01				0.03			
CAD	0.91	-0.23		-0.42	1.57	-0.62		-0.03				0.01				-0.18				0.12			
SEK	1.41	-0.68		-0.09				-0.03				-0.01				-0.04				-0.01			
CHF	0.93	-0.52		0.00				-0.01				0.00				-0.03				0.23	0.06	0.12	0.27
AUD	0.95	-0.31		0.06				0.02				0.02				-0.10				0.03			
CNY	1.33	-0.53		-0.16				-0.01				-0.04	-0.05			0.05				0.00			
INR	0.91	0.04	-0.43	-0.52	0.50			-0.01				0.01				0.26				0.01			
KRW	0.70			-0.11				0.00				-0.02	-0.03	0.00	-0.03	-0.17				0.02			
IDR	0.88	-0.42		0.24				-0.03				0.01				1.32				0.00			
MYR	0.63			0.12				0.00				-0.01	-0.04	-0.01	-0.03	0.29	-0.57			0.03			
THB	1.19	-0.51		0.03				-0.04				-0.01	-0.02			-1.22	-0.20	1.66		0.00			
PLN	0.88			0.03				0.01				0.00				0.31	-0.15	-0.65		-0.04			
TRY	1.02	-0.46		0.47				-0.13				0.01	-0.04			0.90				0.15			
ZAR	1.04	-0.42		0.77				0.02				-0.01	-0.03	-0.01	-0.02	1.39				-0.01			
BRL	1.35	-0.67		1.09				-0.08				-0.02	-0.05			0.51				-0.10			
MXN	1.30	-0.54		0.67				-0.03				0.03				0.50				0.14			
ARS	0.87			-0.31	2.58			0.03				0.01	-0.06			0.42	-0.90	0.19	1.07	-0.04			

Source: Morgan Stanley Research; Note: Shaded cells indicate p-value < 0.05.

Appendix 4: An Overview of Equilibrium Exchange Rate Models

In FX, there are many different frameworks used to derive the fair value of a currency:

It's important to note that there exists no unique or universally accepted concept of an equilibrium exchange rate. Instead, a variety of theoretical and empirical frameworks have been developed, each reflecting different assumptions about macroeconomic balance, policy objectives and market behavior. These models aim to quantify the 'equilibrium' or 'fair value' of an exchange rate as a function of economic fundamentals over varying time horizons. Importantly, they do not predict exchange rate levels with precision but rather provide benchmark estimates to assess misalignments.

Given the high noise-to-signal ratio in FX markets, driven by real shocks, nominal shocks and speculative behavior, these models serve as guiding frameworks rather than deterministic forecasts. As such, they are best interpreted as informative tools for macroeconomic surveillance, rather than prescriptive instruments.

The most commonly used FX fair value models are:

- **Purchasing Power Parity (PPP):** The PPP framework is rooted in the law of one price, asserting that exchange rates should adjust to equalize the prices of identical baskets of goods across countries. In its relative form, it implies that the exchange rate change should mirror the inflation differential between two economies. PPP provides a long-run anchor, often used as a baseline to detect currency overvaluation or undervaluation. However, empirical deviations from PPP are common and persistent, due to price stickiness, non-tradables and cross-border frictions.
- **Capital-Enhanced Equilibrium Exchange Rate (CHEER):** The CHEER model combines the insights of uncovered interest parity (UIP) and purchasing power parity (PPP) into a unified framework, explaining nominal exchange rate movements through both price differentials and interest rate differentials. By integrating goods and financial market equilibria, CHEER captures short-to-medium-run exchange rate behavior driven by capital flows and relative monetary policy stances.
- **Fundamental Equilibrium Exchange Rate (FEER):** The FEER is defined as the real effective exchange rate (REER) that ensures simultaneous attainment of internal balance (full-employment output with low inflation) and external balance (a sustainable current account position). It is a normative, medium-term construct derived from a macroeconomic identity linking the current account to domestic and foreign output, trade elasticities and capital flows. Estimation involves calibrating the sustainable current account norm and solving for the REER that achieves it. While useful for policy simulation, FEER's reliance on assumptions about potential output, trade elasticities and equilibrium capital flows makes it sensitive to parameterization and less suited for short-term dynamics.
 - **Desired Equilibrium Exchange Rate (DEER):** The DEER extends the FEER by embedding explicit policy targets, such as desired current account balances or fiscal positions, into the estimation framework. It retains the structure of the

FEER but introduces a normative layer based on strategic preferences. This allows for scenario-based analysis, estimating what the exchange rate should be under a given policy configuration.

- **Behavioral Equilibrium Exchange Rate (BEER):** The BEER is a reduced-form empirical model that estimates the REER as a function of observed economic fundamentals, such as net foreign assets, terms of trade, relative productivity and interest rate differentials. Unlike FEER or DEER, BEER does not require a current account norm or potential output, making it more flexible and data driven. It captures both short and medium-run effects by including stationary and non-stationary regressors, typically estimated through cointegration techniques. The model's main limitation lies in its theoretical structure and sensitivity to variable selection, especially in structurally evolving economies.
 - **Permanent Equilibrium Exchange Rate (PEER):** The PEER is a structural refinement of the BEER that isolates the long-run (permanent) component of the REER by filtering out temporary or cyclical influences. It replaces actual fundamentals in the BEER model with their trend values, often extracted using statistical filters. As a result, the PEER is smoother and more appropriate for assessing sustained misalignments, particularly in contexts where fundamentals are slow moving or stable.
- **Natural Real Exchange Rate (NATREX):** The NATREX model defines the real exchange rate that would prevail in the absence of speculative pressures and transitory shocks, assuming full employment and long-run equilibrium in capital accumulation and net foreign asset positions. It is rooted in intertemporal macroeconomic theory, where the equilibrium exchange rate evolves dynamically as the economy converges to a steady state defined by exogenous fundamentals (e.g., productivity, social thrift). Unlike FEER and DEER, it does not require predefined policy targets but assumes endogenous convergence towards balance.

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Global Macro Strategy Team

MORGAN STANLEY & CO. LLC	Matthew Hornbach, CMT Matthew.Hornbach@morganstanley.com	Global Head of Macro Strategy	+1 212 761-1837
	Bradley Tian	Global Macro Commentary	+1 212 761-0826
	Martin Tobias, CFA, CMT	US Rates Strategist	+1 212 761-6076
	Aryaman Singh	US Rates Strategist	+1 212 761-1993
	Eli Carter	US Rates Strategist	+1 212 761-4703
MORGAN STANLEY & CO. LLC	Andrew Watrous	G10 FX Strategist	+1 212 761-5287
	Zoë K. Strauss	G10 FX Strategist	+1 212 761-0407
	Molly Nickolin	G10 FX Strategist	+1 212 761-3592
	Simon Waever Simon.Waever@morganstanley.com	Head of EM Sovereign Credit and Latin America Fixed Income Strategy	+1 212 296-8101
	Ioana Zamfir	Head of Latin America Macro Strategy	+1 212 761-4012
	Emma Cerda	Latin America Sovereign Credit	+1 212 761-2344
	Sophia Palacios	Latin America Macro Strategist	+1 212 761-0428
	James K. Lord James.Lord@morganstanley.com	Global Head of FXEM Strategy	+44 20 7677-3254
	Gianluca Salford Luca.Salford@morganstanley.com	Head of European Rates Strategy	+44 20 7677-1337
	Lorenzo Testa	Euro Area Rates Strategist	+44 20 7677-0337
MORGAN STANLEY & CO. INTERNATIONAL PLC	Fabio Bassanin, CFA	UK Rates Strategist	+44 20 7425-1869
	Maria Chiara Russo	Euro Area Rates Strategist	+44 20 7677-3499
	David S. Adams, CFA David.S.Adams@morganstanley.com	Head of G10 FX Strategy	+44 20 7425-3518
	Neville Mandimika	CEEMEA Sovereign Credit Strategist	+44 20 7425-2509
	Arnav Gupta	CEEMEA Rates Strategist	+44 20 7677-0382
	David Cueva Oemer	CEEMEA Rates Strategist	+44 20 7425-2143
	Gek Teng Khoo	AXJ Macro Strategist	+852 3963-0303
	Nimish Prabhune	AXJ Macro Strategist	+91 22 6996-1862
MORGAN STANLEY MUFG SECURITIES CO., LTD.	Koichi Sugisaki Koichi.Sugisaki@morganstanleymufg.com	Head of Japan Macro Strategy	+81 3 6836-8428
	Hiromu Uezato	Japan Macro Strategist	+81 3 6836-8431

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1499	41%	386	46%	26%	699	41%
Equal-weight/Hold	1618	44%	375	44%	23%	778	45%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	581	16%	86	10%	15%	234	14%
Total	3,701		847			1712	

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